NORTHEASTERN UNIVERSITY
IE7615 Deep Learning

Recognising, Finding and Explaining 73 Everyday Objects



Discriminative Project: Final Report (Milestone 4)

Submitted by: Group 7

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October 18, 2026

Abstract

Students in the course photographed 73 everyday objects about a hundred times each, mostly in their own rooms. This report tells the complete story of what Group 7 built from those photographs over four milestones, and how far each result can be trusted. In Milestone 1 five convolutional networks were trained with one shared recipe; transfer learning decided the outcome, and ConvNeXt-Tiny, chosen for robustness by a rule fixed in advance, recognised single photos with 99.23% test accuracy. Milestone 2 asked for every object in grids of 4 to 25 photos. No bounding boxes existed, so 6,628 were made with three box-drawers that fail in different ways, a vote and targeted human checks (accepted test boxes 98.9% right); five YOLO runs and two rounds of label rules produced a YOLO26-s detector with mAP@0.5 0.947. Its explanations, shown in an explorer we built (the EYE), and a diagnosis on validation data exposed what it could not do: rotated photos (recall 0.96 upright, 0.76 to 0.83 rotated), tiny objects and inconsistent labels. Milestone 3 had every one of 6,881 boxes checked by a person, rebuilt the training data with rotations, scattered scenes, SAM cut-outs and empty scenes, and let a rule sealed before the second model's result choose between two retrained models. The selected model boost, improved recall on every test set, each gain with a paired 95% interval excluding zero: rotated grids 0.756 to 0.922, normal grids 0.912 to 0.949, and cut-out scenes 0.368 to 0.881, which revealed how much the earlier model had relied on its photos' own backgrounds.

For this final report every number was re-read from the project files and new measurements were made. They show that about 0.6 recall points of the normal-grid gain came from correcting test labels with the new model's help; that the Milestone 1 classifier still names single photos better than any detector (99.5% against 97.2%); and that the final model is worse than its predecessor on heavily degraded images. A combination of the four best detectors and the classifier, COMPOUND, was pre-registered and tested under the same standard: it raised fully correct validation images by 6.9 points but lost recall on cut-out scenes and added false boxes, so it was rejected as a model and kept as an explanation view in the EYE. Three lessons run through the work: pretrained weights decide first, consistent labels matter more than a better architecture, and only decisions made on validation, by rules fixed before the results, make a number worth reporting.

Keywords: object detection, image classification, transfer learning, YOLO, label quality, data leakage, pre-registration, model ensembles, explainable AI.

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1. Introduction

1.1 The problem

The discriminative project gives each team the same dataset: 73 objects, each photographed about a hundred times by one student. A mouth-guard case, a tissue box, three different watches, two pairs of sunglasses, a table-tennis paddle beside a tennis racket, a keyboard, an apple. The photographs are ordinary phone pictures, taken near and far, upright and at angles, in good and poor light, against the beds, desks and floors of student rooms. The question the course asks becomes harder at every milestone: first *which* object a photo shows, then *where* every object is in an image that contains up to 25 of them, and finally whether that still works when the image looks nothing like the training data.

Two features of the data shape everything that follows. Each object was photographed by one person in one set of rooms, so the object and its background travel together; a model can learn the room instead of the object. And the data came with names but no boxes: a detector needs a box around every object both to learn and to be graded, and making about 7,000 trustworthy boxes became the largest single task of the project.

1.2 The story in brief

Each chapter opens with a map of its part of the project (Figure 1, Figure 6, Figure 8, Figure 14 and Figure 22); read in order, the five maps form one flowchart of the whole work. The lanes separate what the models learned from, where people decided, what was trained, and how results were judged; green cards mark decisions made by people and selected results, dashed cards mark findings, and each card carries its real numbers. The thread running through them is a cycle that repeated three times: build something, measure it honestly, and let the measurement decide what to fix next.

Milestone 1 built a classifier and measured it under simulated photo problems; the measurement chose the model and also revealed that every network partly recognised the room. Milestone 2 had to invent its own boxes before it could train a detector; the detector was good on the grids it was trained for, and the EYE, our explorer, showed exactly where it failed. Milestone 3 measured those failures on validation data before changing anything, repaired the labels at their source, rebuilt the training data around the failures, and chose the new model by a rule sealed before the deciding result. This report closes the cycle a last time: it re-reads every number from the files, measures what the earlier reports could not, and tests whether combining the best models helps.

1.3 What this report adds

The milestone reports [20] were written under deadline, one milestone at a time. This report replaces them as the single record of the project. Every number was re-read from the project's result files into a fact sheet with sources, and where the earlier reports were wrong or incomplete the correction is stated with its evidence (Appendix E). New measurements put 95% intervals and paired tests on every final result, quantify a bias in how the final test labels were corrected, compare the detector's naming with the classifier on identical photos, and test degraded images and empty backgrounds (Section 8). A combination of the best models was tested as a pre-registered experiment (Section 9). The discussion follows four threads through all milestones and states the threats to validity plainly (Section 11).

1.4 Setting and conventions

Group 7 is Tanmayi Shurpali, Santhoshkumar Chandrasekar, Praniti Sunil Kale and Mukul Sridar Haladhi; the instructor is Xuemin Jin. All members sent their code, data and results to one integrator, Tanmayi, who committed them to the shared repository, so that one copy of the code and one dataset state existed; every

commit therefore carries one name although the work was shared. Training ran on one laptop GPU (NVIDIA RTX 5070, 8 GB) with Python 3.14.2, PyTorch 2.11.0 and Ultralytics 8.4.174 throughout.

Some numbers legitimately differ for the same model, because the answer key or the scoring code changed between milestones. Every result in this report therefore names its truth version (Table 1), its scorer and its settings. Three scorers appear: Ultralytics' validator in Milestone 2, the cross-evaluation code of Milestone 2, and the project's scoring code from Milestone 3 onwards. On the 1,041 Milestone 2 test objects they give the selected Milestone 2 model mAP@0.5 0.947, 0.943 and 0.940. Unless a table says otherwise, grids are scored at confidence 0.10 with one box per 224-pixel cell, and scenes at 0.35 after class-agnostic duplicate removal. Intervals are 95% bootstrap intervals over images (1,000 resamples, seed 0), and differences between two models use a paired bootstrap on the same draws. Numbers marked * are derived by subtraction.

Table 1. The three answer keys of the test set.

Key	What it is	Normal-grid objects	Used for
T1	the Milestone 2 test grids and boxes	1,041	Milestone 2 results; the bias-free comparison
T2	after pass 1 of the Milestone 3 label check	1,044	boost's own evaluation
T3	after pass 2: the best available truth	1,048	final results, intervals, COMPOUND

T2 and T3 add second objects visible in a test photo; T3 also ignores one test photo excluded in pass 2.

2. The Data

A model can only be as good as the evidence it is trained and judged on, so the project began with the data, not the models. This chapter follows the photographs from the course folders to the split that served all milestones, and explains how the work was protected from the most common way evaluations go wrong: leakage between training and test data.

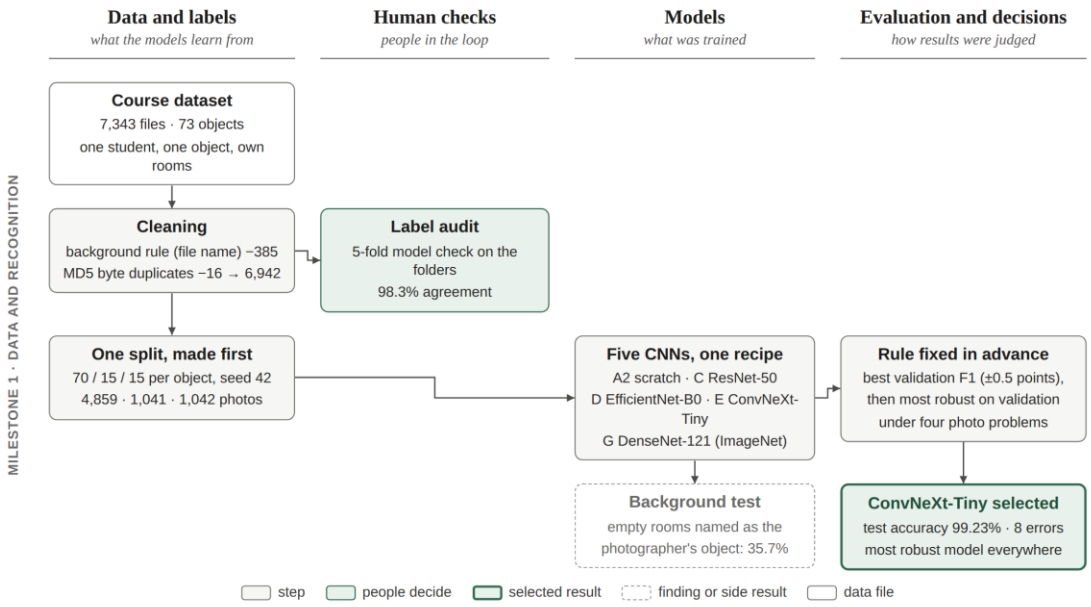


Figure 1. *Map of Milestone 1: from the course folders to the selected classifier (Chapters 2 and 3).*

2.1 Collection, and the bias built into it

The course dataset holds 7,343 image files in 73 object folders of 100 to 113 files. Some folders also contain *background-only* photographs: the empty room, without the object. Because each object was photographed by one student, mostly in their own rooms, object and background are correlated by construction. This is not a flaw anyone could clean away; it is the main bias of the data, and the project measured its effect in every milestone (Section 11).

2.2 Cleaning

Table 2. Cleaning in Milestone 1.

Step	Photos left	How it was done
Index all files	7,343	73 classes; no unreadable files
Remove background-only photos	6,958	by file name: `_bg_` or `BG` followed by digits, any case; 385 removed. A CLIP-based detector exists in the notebook but was switched off
Remove exact duplicates	6,942	MD5 hash of each file's bytes; 28 identical groups existed before the background step, 16 copies were removed after it; the copy whose name sorts first is kept
Split 70/15/15	4,859 / 1,041 / 1,042	stratified by object, seed 42; 62–74 training, 14–16 validation and 13–16 test photos per object

After cleaning each object has 89 to 106 photos (median 95). A fixed seed makes the shuffle, and therefore the split, identical whenever it is rerun.

Both rules were simple and logged, and both had gaps that later milestones closed. The background rule trusts file names, so background photos a student did not tag stayed in their class: five OBJ004 photos show the room without the toy car. Milestone 2 removed them by eye before any box was made, and Milestone 3's full label check excluded 14 more photos in which no object can be identified (Figure 17). The duplicate rule compares bytes, so a re-saved copy or a second shot taken a moment later is not caught; Milestone 1 measured these near-twins (99 test photos have one in training), and Milestone 3 found 183 near-identical pairs across the splits and removed 41 training and validation copies. One small inconsistency is also explained here: Milestone 1's background test used 378 photos although 385 were removed. No file records why; the most likely reason is that the 385 contain seven byte-identical copies (385 − 7 = 378) and the Colab notebook removed duplicates before filtering the backgrounds.

2.3 One split for the whole project

The split was made once, before any box or model existed, and kept for all milestones: a test photo in Milestone 1 is a test photo in Milestone 3. Every grid, scene and cut-out was later built only from photos of one split. Figure 2 follows the photos through the project; the small differences between stages are the exclusions described above.

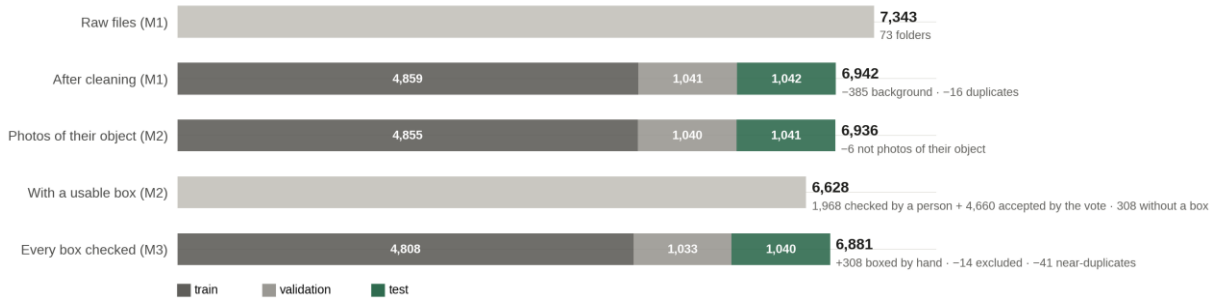


Figure 2. *From 7,343 files to 6,881 photos with a person-checked box, with the split at each stage.*

Milestone 1 also audited the folder labels: a logistic regression on frozen ConvNeXt-Tiny features, cross-validated in five folds, agreed with the folder for 98.3% of photos in the fully logged run (97.9% in the submitted study), a high level for 73 classes with no sign of systematically mislabelled folders.

2.4 Guarding against leakage

Kapoor and Narayanan [7] identify leakage as the most common reason machine-learning results fail to hold: training and test data that are not cleanly separated, features that should not be available, and test data that do not represent the intended use. Table 3 lists how each applies here and what was done. Two residual risks could not be designed away; they are measured in Section 8 and discussed in Section 11.

Table 3. Leakage risks and controls across the project.

Risk	Control	What remains
Train/test overlap	one split by photo, made first; images built from one split only	none found
Duplicates across splits	MD5 de-duplication before the split; 41 near-duplicates removed in Milestone 3, always keeping the test photo	the Milestone 1 classifier trained before that removal
Shortcut through the room	background test; unseen test backgrounds and cut-outs in Milestone 3; empty-background test	a measured, weak cue
Tuning on the test set	every choice on validation; two rules sealed with SHA-256 before their results	the same test photos served three milestones
Answer key shaped by the model	people decided every label change; all models scored on the same truth	a measured bias of 0.6 recall points
Teammates' runs trained on our test photos	those objects removed in the cross-evaluation; the runs kept out of COMPOUND	none
Test data unlike the intended use	one test set per situation: normal, rotated, scattered, cut-out, single photos, degraded	scattered and cut-out scenes are synthetic

3. Recognising One Object

Milestone 1 asked the simplest form of the question: given one photo, which of the 73 objects is it? The aim was not only a high score but an understanding of what drives it, so the comparison was designed to isolate the architecture, and the choice between models was decided by a rule written before the test set was used.

3.1 Five networks, one recipe

A custom VGG-style CNN with ten convolutional layers, trained from scratch, was compared with four ImageNet-pretrained networks of different design: ResNet-50 [4], EfficientNet-B0 [17], ConvNeXt-Tiny [9] and DenseNet-121 [5], shortlisted from exploratory runs by two members on earlier cleanings. All five used one recipe so that differences reflect architecture rather than tricks: AdamW [10] without weight decay on biases and normalisation layers, a new output layer learning 3 to 10 times faster than pretrained layers, warm-up then cosine decay, label smoothing 0.1 with class weights, mixed precision, gradient clipping, and early stopping and checkpoint selection on validation macro-F1.

Transfer learning was decisive: every pretrained network beat the scratch CNN by about ten points. Among the four pretrained models, however, the differences on clean photos lie inside each other's intervals, and McNemar's test [11] on the same photos confirms that ConvNeXt-Tiny is not significantly better than the others there. A choice based on clean accuracy alone would have been a choice based on noise.

Table 4. Milestone 1 on the 1,042 test photos (submitted main study).

Model	Val F1	Test accuracy (95% CI)	Hard subset	Errors	Params	Train time
A2 Custom CNN (scratch)	89.56	89.06 (87.1–91.1)	87.91	114	3.41 M	30.6 min
C ResNet-50	99.22	98.66 (98.0–99.3)	98.52	14	23.66 M	11.4 min
D EfficientNet-B0	99.41	98.75 (98.1–99.3)	98.62	13	4.10 M	11.7 min
E ConvNeXt-Tiny (selected)	**99.61**	**99.23 (98.7–99.7)**	**99.15**	**8**	27.88 M	13.4 min
G DenseNet-121	99.13	98.66 (97.9–99.3)	98.52	14	7.03 M	14.1 min

Hard subset: the 943 test photos without a near-twin in training. A fully logged local run with the same split also selected E (99.42%).

3.2 Choosing for robustness

Real photos are often small, cropped, partly hidden or poor. Four such problems were simulated on every validation and test photo (Figure 3): a small object (half size on a blurred canvas), a 70% crop, a grey box over a quarter of the image, and low quality (4× downscale, brightness × 0.6, JPEG quality 20). The rule, fixed before testing, kept the models within 0.5 points of the best validation F1 and then chose the most robust on *validation* perturbations. ConvNeXt-Tiny led on validation (83.98%) and, as Figure 4 shows, on test too. Low-quality photos were the hardest condition for every model (best 51.7%), a weakness that returns in every later milestone.



Figure 3. *The four simulated photo problems applied to one test photo: original, small, cropped, occluded and low quality.*

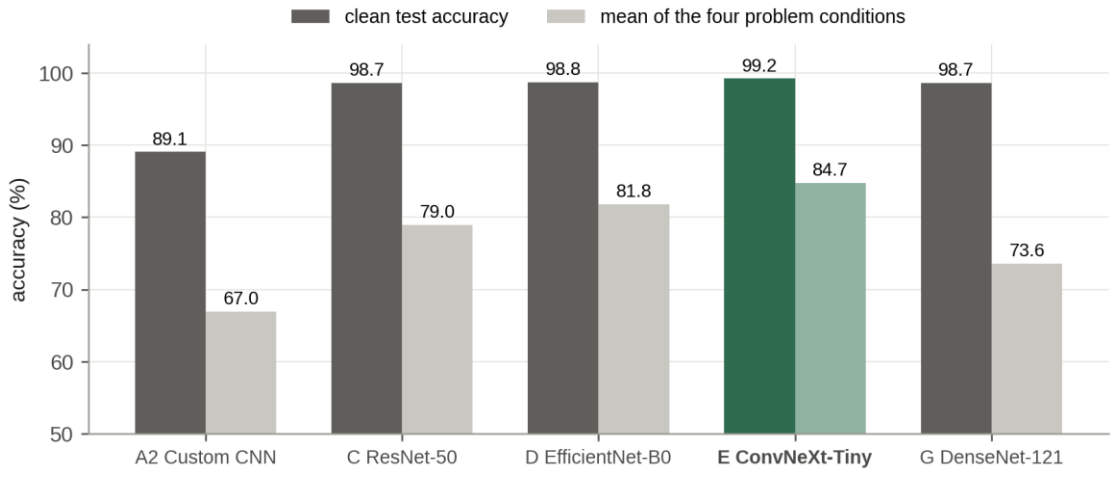


Figure 4. *Clean accuracy against robustness. On clean photos the pretrained models are tied; under the four simulated problems ConvNeXt-Tiny is clearly the best.*

3.3 What the classifier looks at

Grad-CAM [15] heat maps (Figure 5) show the pretrained networks focusing tightly on visible objects. A sharper test asked what the models do when there is *no* object: shown photos of the empty rooms, every model named the photographer's own object for about a third of them (ConvNeXt-Tiny 35.7%), roughly 25 times chance, although with low confidence (0.36 against 0.89 on real photos). The room is a weak secondary cue. Milestone 1 could only measure it; the detectors of the later milestones would show how much it mattered.

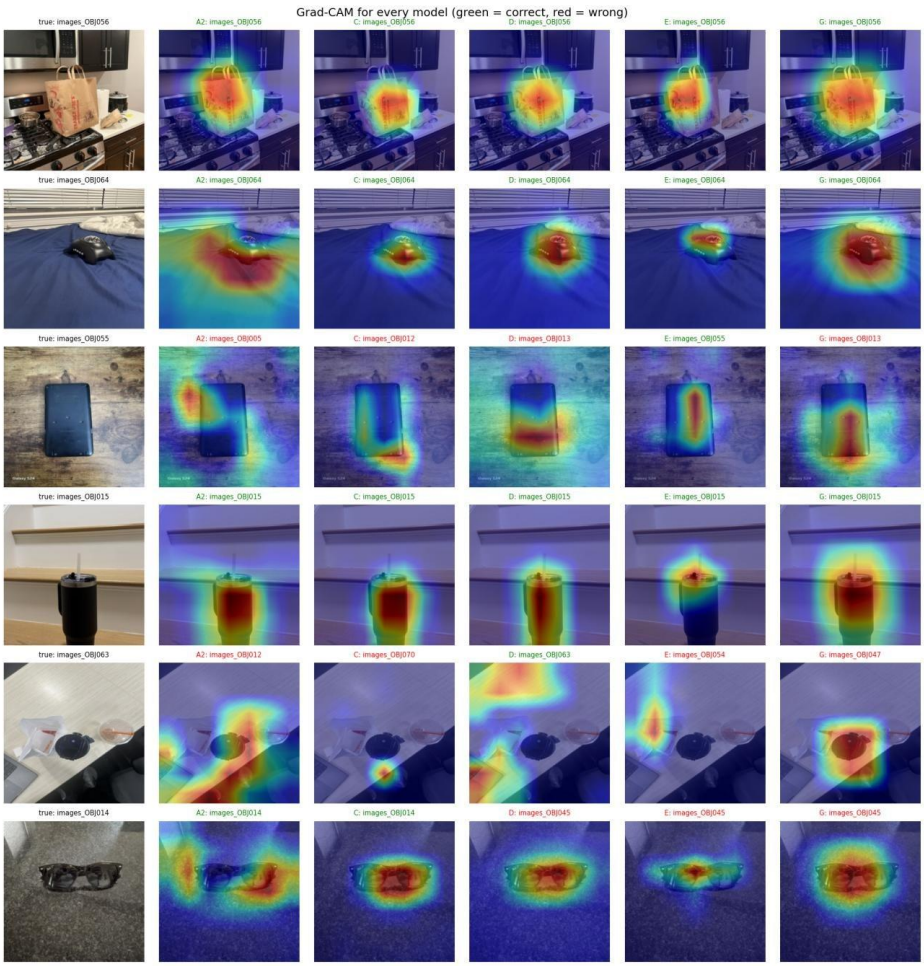


Figure 5. *Grad-CAM heat maps for six test photos across the five models (green title correct, red wrong).*

One limitation of the record should be stated here. The submitted main study ran in Google Colab and its notebook was saved without outputs, so its numbers come from the Milestone 1 report alone. The local run with the same split, recipe and rule is fully logged and reached the same conclusion; we keep 99.23% as the official result and the local run as a replication.

4. Making the Boxes

Milestone 2 changed the question from *which* to *where*: find and name every object in a grid of 4 to 25 photos. A detector learns from boxes and is graded against boxes, and the dataset had none. Drawing about 7,000 by hand was impractical, but machine-made boxes are only useful if their accuracy is known. The design followed three principles set before starting: trust a box only when methods that fail in different

ways agree on it; let people check every test box, because the test set produces every reported number; and measure the quality of the machine boxes rather than assume it.

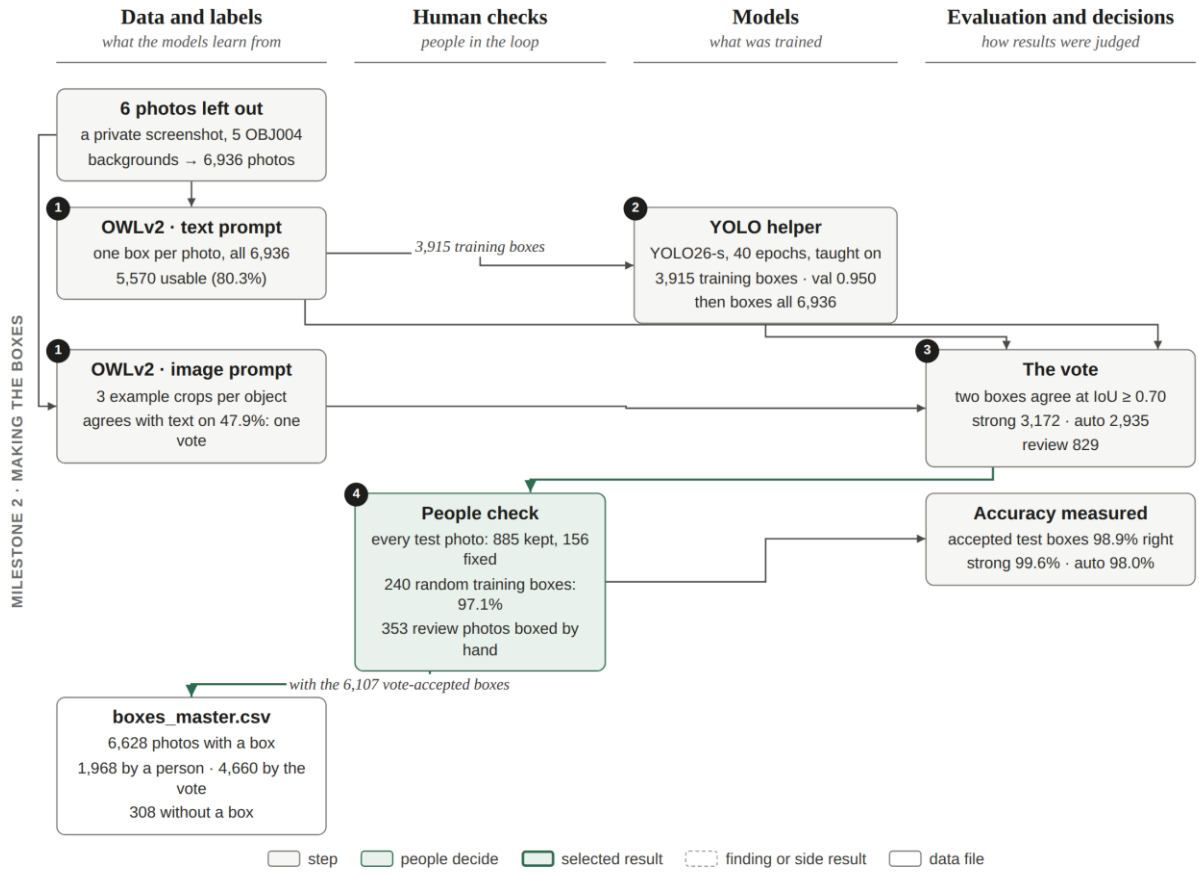


Figure 6. *Map of the box pipeline in Milestone 2. Numbers 1–4 give the order: two OWLv2 prompts and the trained helper each box every photo, the vote combines them, and people check where it matters.*

4.1 Three box-drawers and a vote

The three box-drawers are steps 1–3 in Figure 6. Two use OWLv2 [12], an open-vocabulary detector that finds whatever a sentence or an example image describes: once with the object's name as a text prompt ("a photo of a toy car"), and once with three example crops per object. The third is a YOLO26-s helper we trained ourselves, for 40 epochs, on the 3,915 usable text-prompt boxes of training photos only; it reached validation mAP@0.5 0.950 on single photos, and because it never saw a validation or test photo, its opinion there is independent.

Table 5. What each box-drawer contributed (6,936 photos).

Drawer	Behaviour	Usable result
OWLv2 text prompt	ran on all photos in about 17 minutes after a GPU-memory fix (16× faster); flagged boxes with score below 0.15 (1,206), covering under 1% of the photo (144) or over 97% (16)	5,570 boxes (80.3%)
OWLv2 image prompt	often boxed floor, bed or table; agreed with the text prompt on only 47.9% of photos	one vote, never trusted alone
YOLO helper	trained on training photos only, then boxed every photo	an independent third opinion

The vote counts two boxes as agreeing when their IoU (intersection over union) is at least 0.70; the helper may vote only at confidence 0.25 or more, and every voting box must cover 1–97% of the photo. All three agreeing gives a *strong* box (3,172 photos), exactly two an *auto* box (2,935), and anything else goes to people for review (829: 794 with no agreement and 35 where the helper confidently saw another object).

4.2 People in the loop, and when

Figure 7 follows the vote's output by set. Every one of the 1,041 test photos was checked right after the vote, which also measured the vote: 863 of the 873 accepted test boxes were right (98.9%; strong 99.6%, auto 98.0%), and the photos that needed fixing had a median IoU of only 0.23 between the automatic and the human box, so the vote's failures were clear failures rather than slightly misplaced boxes. A random sample of 240 accepted training boxes was 97.1% right. On that evidence the remaining 4,660 accepted training and validation boxes were trusted without a person seeing them, a decision Milestone 3 would revisit.

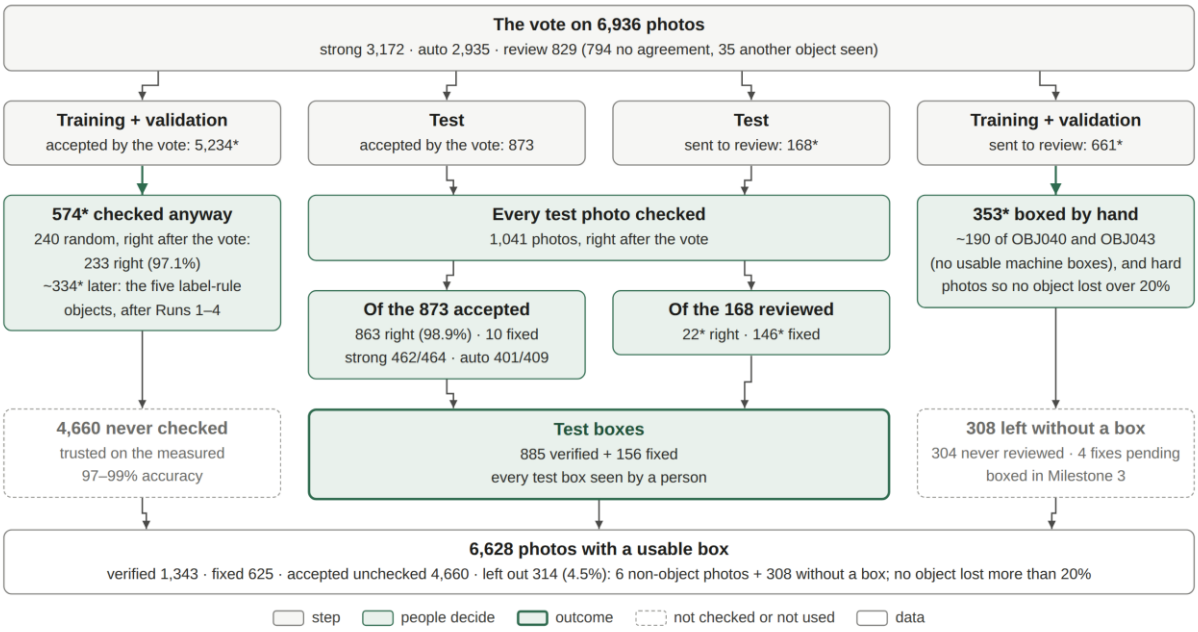


Figure 7. *Where every box came from. The 574 training and validation boxes a person saw were checked at two different times: 240 right after the vote, and about 334 after training Runs 1–4 had revealed the five label-rule objects (Section 5.2).*

The box file records one status per photo: verified 1,343, fixed 625, accepted unchecked 4,660, and 308 without a usable box (304 never reviewed, 4 with a fix pending). That gives 6,628 photos with a box. The 314 left out (4.5%) were spread so that no object lost more than 20% of its photos.

5. The First Detector

With boxes in hand, Milestone 2 built the multi-object images, trained five detectors, and found that the labels, not the architecture, were the main lever. This chapter also reports what the selected detector could and could not do, because its failures set the agenda for Milestone 3.

5.1 From photos to grids

Each grid places 224-pixel photos in a 2x2 to 5x5 arrangement and moves each box with its photo: a box from (50, 30) to (180, 200) placed in column 2, row 1 becomes (498, 254) to (628, 424). Grids were built only from photos of one split (Figure 9). The test set has 72 grids (16 of 2x2 and 14 of each other size) with 1,041 objects, its 2x2 grids holding 61 objects and three empty filler cells; validation has 68 grids with 972 objects. Training used 1,500 images, built once per dataset version and fixed on disk: grids only, 300 per size, for Runs 1–4, and for Run 5 85% grids (half with look-alike objects side by side) plus 15% single photos.

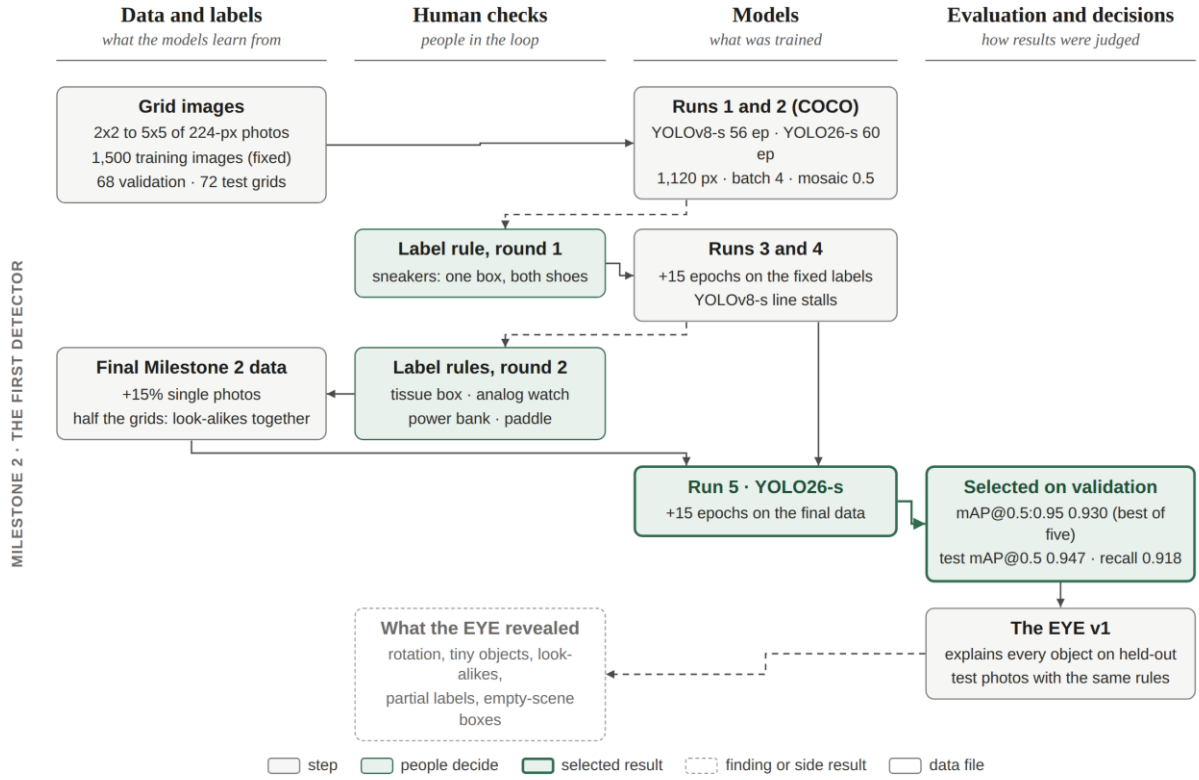


Figure 8. *Map of the first detector: grids, five runs, two rounds of label rules, the validation choice and the EYE.*

The 1,500 images reuse the same training photos: in Runs 1–4 their 22,200 boxes come from 4,615 training photos, so each photo appears about five times, in different neighbourhoods: like reshuffling one deck of cards, there are no new cards but many new hands. Two layers of augmentation add variety. About 20% of images were baked with blur, low resolution, JPEG artefacts or darkness when built, and Ultralytics changes every image each time it is loaded: mosaic of four images (probability 0.5, switched off for the final epochs), small changes of hue, saturation and brightness, left–right flips, scaling and shifts, but no rotation. Because Ultralytics resizes every image to the 1,120-pixel input, an object appears at different sizes depending on its grid; validation therefore chose to detect small images at their natural size, and Milestone 3 removed the mismatch altogether.

5.2 Five runs and two rounds of label rules

Two COCO-pretrained [8] detectors were compared: YOLOv8-s, the model named in the brief, and YOLO26-s, the newest small Ultralytics model [6]. All runs used a 1,120-pixel input and batch 4. Runs 1

and 2 trained from COCO; the later runs fine-tuned the previous one at learning rate 0.002. Table 6 lists all five, re-scored on the same final validation and test grids.

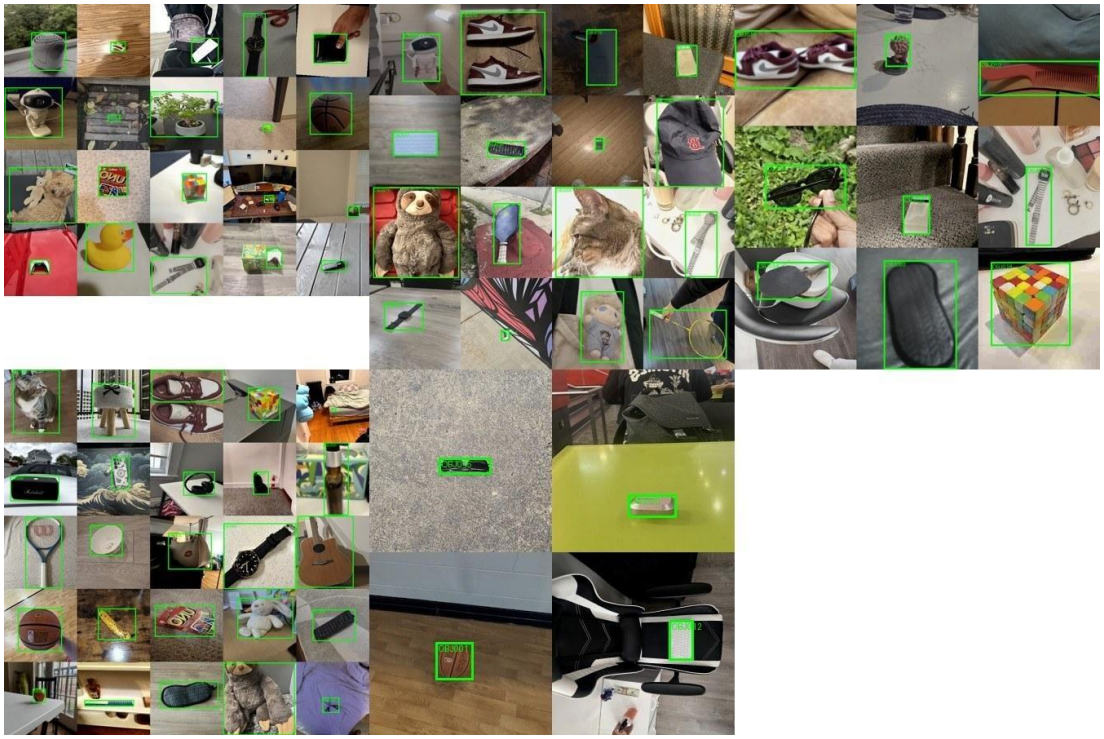


Figure 9. *Test grids of 4, 9 and 16 photos with their boxes.*

Table 6. The five Milestone 2 runs (T1, Ultralytics validator).

Run	Model	Start	Training data	Val mAP@0.5: 0.95	Test mAP@0.5 / 0.5:0.95	Test precision / recall
1	YOLOv8-s	COCO, 56 epochs	grids only	0.897	0.910 / 0.846	0.914 / 0.890
2	YOLO26-s	COCO, 60 epochs	grids only	0.907	0.921 / 0.858	0.922 / 0.884
3	YOLOv8-s	Run 1 + 15 epochs	+ sneaker rule	0.906	0.932 / 0.859	0.925 / 0.894
4	YOLO26-s	Run 2 + 15 epochs	+ sneaker rule	0.919	0.931 / 0.863	0.922 / 0.910
5	**YOLO26-s**	Run 4 + 15 epochs	+ four rules, singles, look-alikes	**0.930**	**0.947 / 0.875**	**0.931 / 0.918**

Runs 1 and 2 took about 70 and 104 minutes. YOLO26-s was trained first; the numbering follows the Milestone 2 report.

After each pair of runs a few objects scored far below the rest. Looking at all of their boxes side by side showed that they had been boxed in two ways (Figure 10): around the tissue paper or the whole tissue box, around the watch case or the case and its strap. A detector trained on such labels receives contradictory answers, and even a perfect prediction is marked wrong against half the answer key. One rule was written per object and every photo of it re-checked in every split, in two rounds: the sneakers after Runs 1–2, then the tissue box, analog watch, power bank and paddle after Runs 3–4 (Table 7, Figure 11).

The four new rules, together with single photos and look-alike grids in the training data, produced the largest single gain of the run history (Figure 12). Run 5 had the highest validation mAP@0.5:0.95 (0.930) and was selected by that rule; the YOLOv8-s line had stalled at Run 3.

(a) OBJ051 tissue, before correction: some boxes cover the whole box, others only the tissue paper



(b) OBJ065 analog watch, before correction: whole watch and strap (often image-sized) or only the metal case

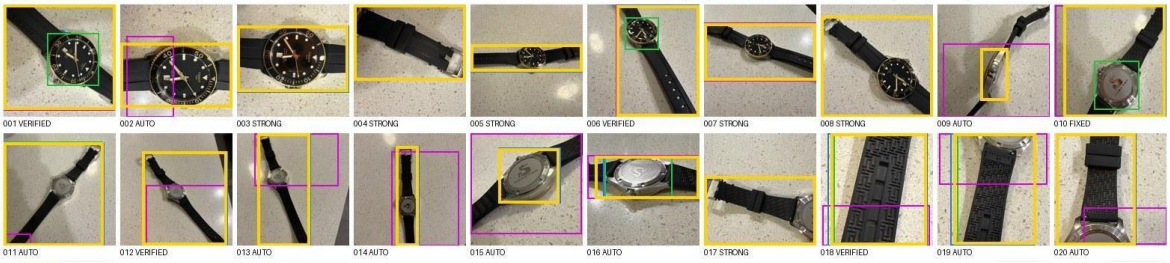


Figure 10. *Two ways of boxing the same object before the rules: the whole tissue box or only the paper (top), case and strap or only the case (bottom).*

Table 7. The five label rules.

Object	Rule	Round	AP@0.5
OBJ010 sneakers	one box around both shoes	after Runs 1–2	–
OBJ051 tissue box	the whole tissue box	after Runs 3–4	0.30 → 0.995
OBJ048 power bank	the silver one only	after Runs 3–4	0.71 → 0.95
OBJ065 analog watch	case and strap	after Runs 3–4	0.59 → 0.81
OBJ034 paddle	the paddle only, never the tennis racket	after Runs 3–4	0.67 → 0.59

AP values as given in the Milestone 2 report; the run and split they were measured on are not recorded.

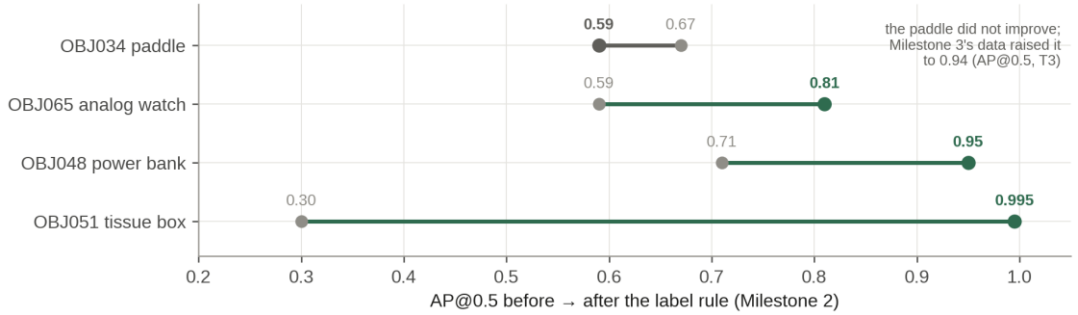


Figure 11. *Three rules lifted their object sharply; the paddle did not improve until Milestone 3's new data (AP@0.5 0.58 → 0.94 on T3).*

5.3 What the first detector could and could not do

On the 72 test grids Run 5 reached mAP@0.5 0.947 (mAP@0.5:0.95 0.875), precision 0.931 and recall 0.918, with 48 of the 73 objects at AP@0.5 0.95 or above. A second test set of 84 images of other kinds included 14 single photos, all correct. Of the 85 objects it got wrong, 69 were missed and 16 named wrongly: the detector mostly knew what things were; its weakness was noticing them. Because one error spoils a whole grid, only 34.7% of grids were fully correct (95% CI 23.6–45.8%), falling from 87.5% for 2x2 grids to 7.1% for 5x5 (Figure 13).

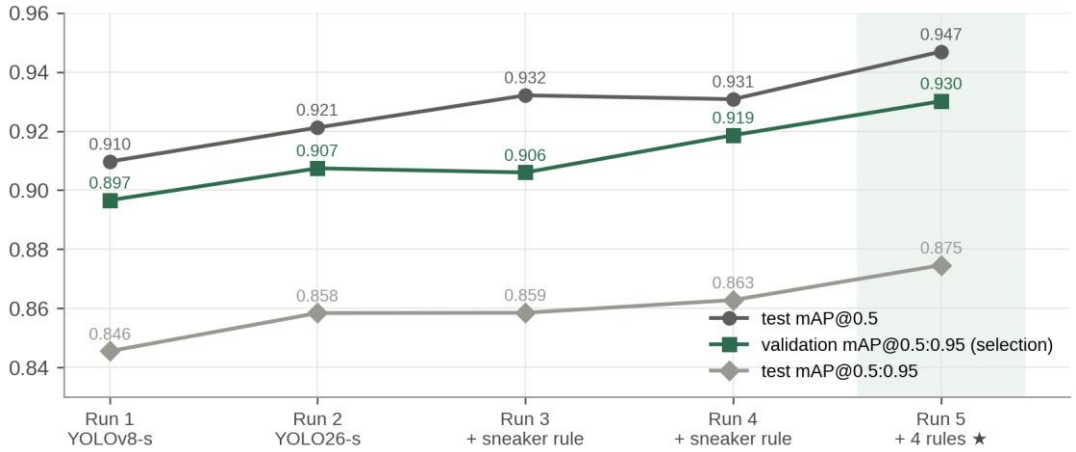


Figure 12. *Scores along the five runs. The YOLO26-s line (Runs 2, 4, 5) improved at every step; the YOLOv8-s line did not (Run 3 validation 0.906 against Run 2's 0.907).*

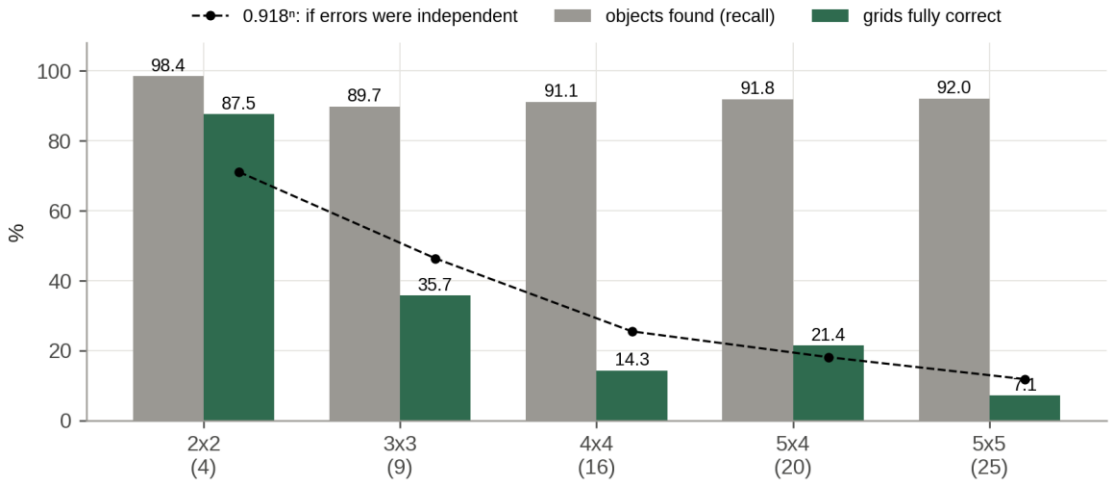


Figure 13. *Objects found stay at 90–98% for every grid size, but one miss spoils a grid; the dashed line shows what independent errors at 91.8% per object would give.*

The weakest objects were look-alikes (paddle 0.59, card wallet 0.62, sunglasses 0.79) and a small object (mouth-guard case 0.77). Inspecting individual failures in the EYE (Section 10) gave them causes: tiny objects (one pencil case covered 1.3% of its photo), confidence just below the threshold (a mouth-guard case seen at 0.0996 against 0.10), look-alikes, partial labels (a watch box "off" at IoU 0.37 because its label covered only part of the watch), weak boxes on backgrounds, and co-occurring objects that often appear in the same photos. Two situations had never been in the training data at all: a photo detected upright at confidence 0.54 was missed when turned 90°, and on heavily degraded grids (¼ resolution, dark, JPEG 20) mAP@0.5 fell to 0.428. These observations became the brief for Milestone 3.

5.4 The teammates' pipelines

Three alternative pipelines built by teammates with other labelling tools, models and splits (runs A–C) were compared with the selected model by the same scoring code on the human-checked test set. Objects a pipeline could not fairly be judged on were removed, with any prediction in their cell: objects of classes it cannot predict, and objects whose photo it had trained on. The selected model was then scored on exactly the same remaining objects. As a check, this code gives the selected model mAP@0.5 0.943 on all 1,041 objects, within half a point of the Ultralytics value.

Table 8. Cross-evaluation on the human-checked test set: each pipeline against the selected model on the same objects.

Run	Objects	Its mAP@0.5	Its mAP@0.5:0.95	Its fully correct	Selected mAP@0.5	Selected fully correct
A	1,019	0.883	0.782	11.1%	**0.944**	**34.7%**
B	628	0.779	0.697	5.7%	**0.923**	**38.6%**
C	543	0.819	0.720	11.3%	**0.949**	**52.1%**

Precision-based measures use each pipeline's own confidence threshold (0.30, 0.15, 0.25; selected 0.10).

6. Diagnosis and Repair

Milestone 3 asked whether the detector could find objects anywhere: rotated, scattered over unfamiliar backgrounds, or lifted out of their photos. The tempting response to the list of failures was to retrain at once. Instead the failures were measured first, on validation data only, so that each fix would answer a measured problem and the test set would stay untouched until the end.

6.1 Measuring before fixing

On upright validation grids the Milestone 2 model was already strong: recall 0.961, precision 0.972, five wrong IDs and 61.8% fully correct grids. Turned by 90°, 180° and 270°, recall fell to 0.771, 0.830 and 0.762 and wrong IDs rose fourteen-fold (Figure 15). Half-turns hurt less, probably because an upside-down object keeps its outline proportions while a sideways one does not. Each of the 38 upright misses was given a cause. The largest group was not a failure to see: 11 objects were named correctly but their box overlapped the truth by less than half, often because the label itself was partial.

Three cheap fixes were tried before any retraining. Filling empty grid cells with their best candidate gained 0.4 recall points but cost 0.7 points of precision; test-time augmentation lowered fully correct grids from 62% to 54%; running the model at 1,280 or 1,600 pixels without retraining made it worse, because it had learned objects at the 1,120 size. All three were rejected, and the decisions followed from the measurements: retrain at 1,280 pixels so small objects are learned at that size, teach rotation with rotated photos, and repair the labels at their source by having a person check every box.

6.2 Every box checked by a person

Checking all ~6,900 boxes in days required a tool that put each photo in front of a person where it most needed attention, and that could not damage the original labels. `m2_label_check.py` (Figure 16) runs a model over every photo as a second opinion and sorts the photos into queues; finds near-identical photos across splits with perceptual hashes; lets people review each queue in a browser, saving every click at once; and writes the result to a new file, never touching the Milestone 2 box file. One rule replaced the five object rules: box the whole object, every visible part, tight to its edges.

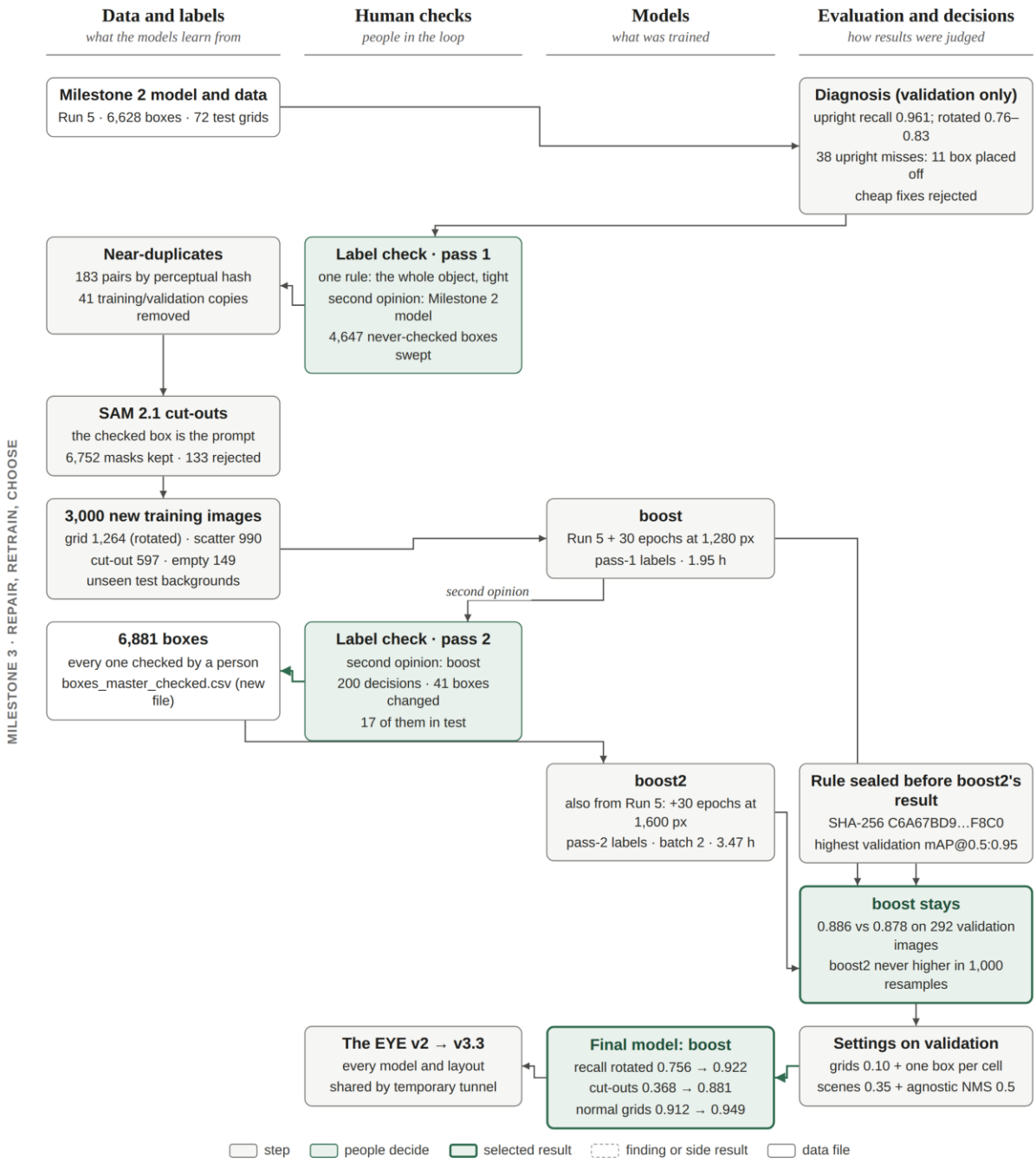


Figure 14. *Map of Milestone 3: diagnosis, the label check in two passes, the new data, two retrained models from the same start, the sealed choice and the final model (Chapters 6 and 7).*

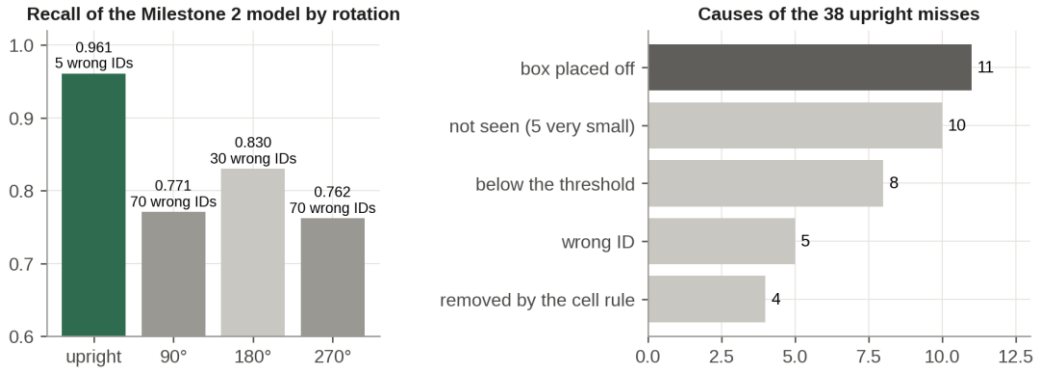


Figure 15. *Diagnosis of the Milestone 2 model on validation: recall by rotation (left) and the causes of the 38 upright misses (right).*

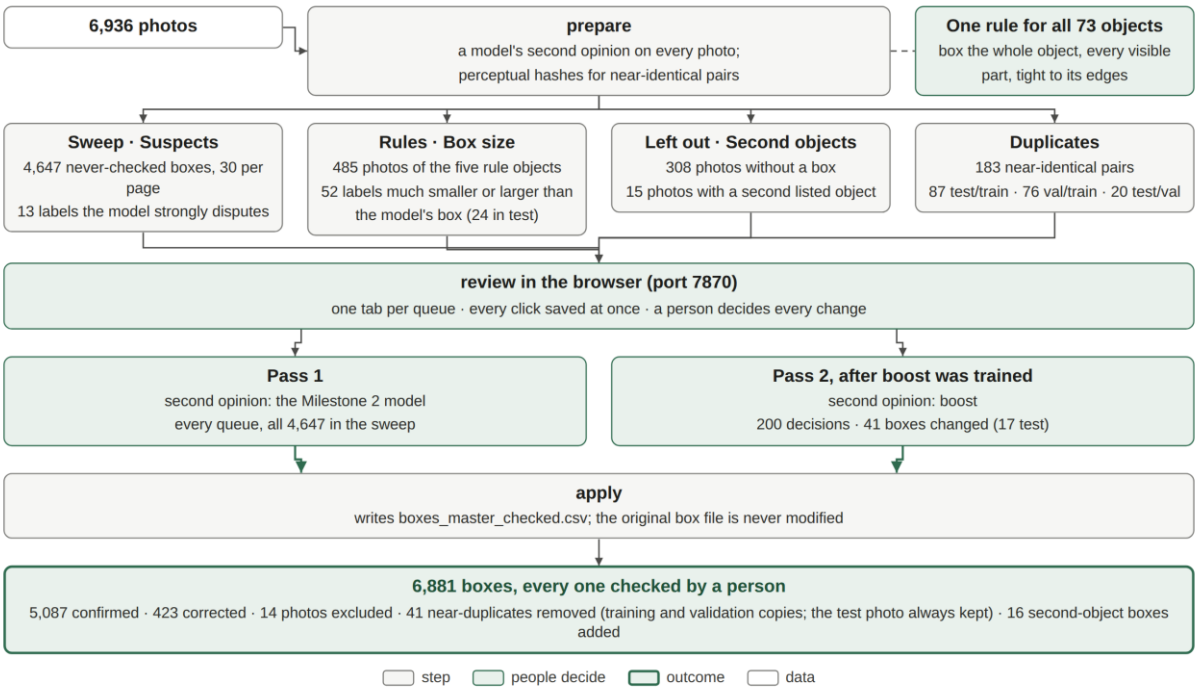


Figure 16. *The label check, from 6,936 photos to 6,881 boxes that a person has seen.*

The first pass used the Milestone 2 model as second opinion and covered every queue, including the 4,647 boxes no person had seen and the 308 photos without a box. A second pass, run after the first retrained model existed, used that model, boost, as second opinion: 200 decisions and 41 changed boxes, 17 of them in the test set. Changing test boxes changes the answer key; how much this favoured boost is measured in Section 8. The 14 photos a person excluded close the background-photo story that began with Milestone 1's filename rule (Figure 17): ten show no object and four only an unidentifiable fragment.

6.3 Teaching rotation, scenes and unfamiliar backgrounds

The new training data was built around the diagnosis (Figure 18). Grids now include photos turned 90°, 180° and 270° with exactly rotated boxes, small tilts with re-fitted boxes and whole rotated grids. Scattered scenes place photos at random sizes and angles on real backgrounds, giving the 385 background photos removed in Milestone 1 a use; the backgrounds were split by a hash of their file names so that test scenes use backgrounds the model never saw. Cut-outs lift each object out of its photo with SAM 2.1 [14],

prompted by the person-checked box, and paste it anywhere: the box stays the checked box, and a quality gate on fill, spill and fit rejected 133 of 6,885 masks. Empty backgrounds teach the model to say "nothing here". Every image is built at the 1,120-pixel scale and only then scaled to the model's input, so an object keeps the same relative size in training, validation and the EYE.

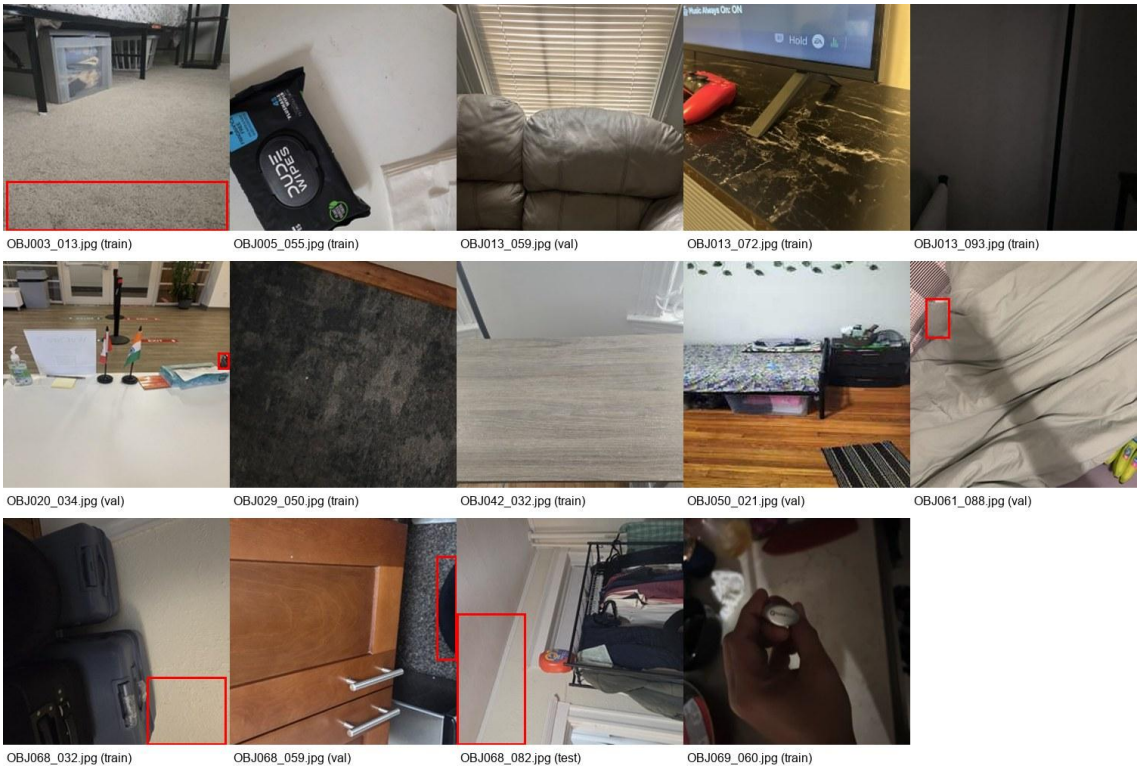


Figure 17. *The 14 photos excluded in the label check, with their recorded boxes in red. None clearly shows its object.*

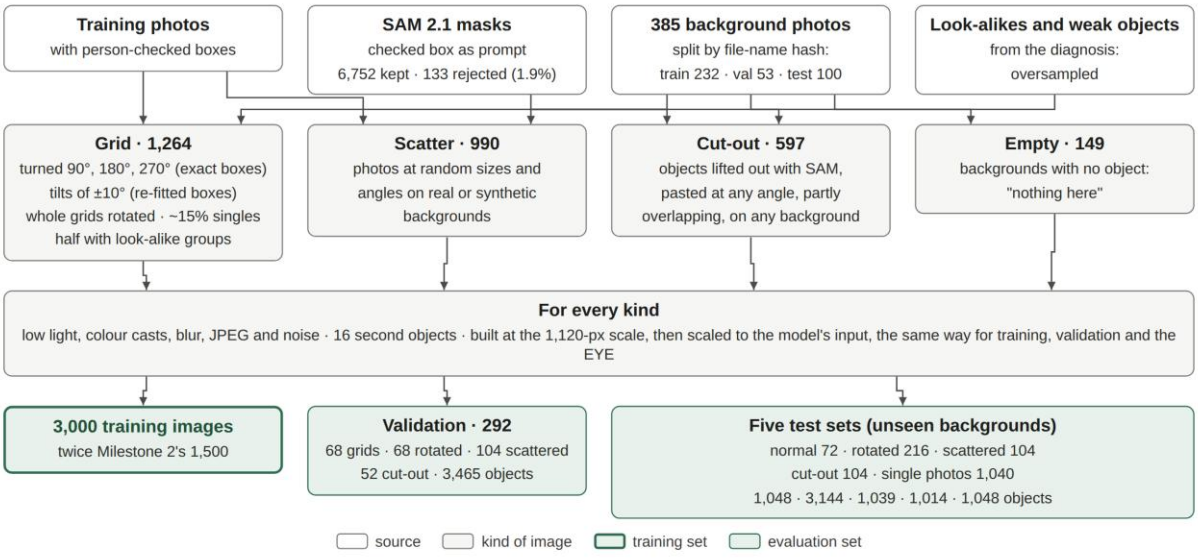


Figure 18. *How Milestone 3's 3,000 training images and its validation and test sets were built. Counts under the test sets are objects on T3.*

7. Retraining and Choosing

Two models were trained from the Milestone 2 model, and the choice between them was taken out of our hands before the second model's result was known. This chapter reports that choice, the final results with their uncertainty, and why training stopped there.

7.1 Two models from one starting point

Both models fine-tune Run 5 for 30 epochs with gentle settings: learning rate 0.002 with cosine decay, patience 10, and mosaic switched off because the new images are already composites. *boost* trains at 1,280 pixels on the pass-1 labels (batch 4, 1.95 hours). *boost2* tests whether more magnification pays: at 1,600 pixels an object covering 1.1% of its photo grows from about 23 to 33 pixels. It trains on the pass-2 labels with batch 2, after a batch-4 attempt was abandoned in its first epoch, and took 3.47 hours. The two validation curves track each other closely (Figure 19).

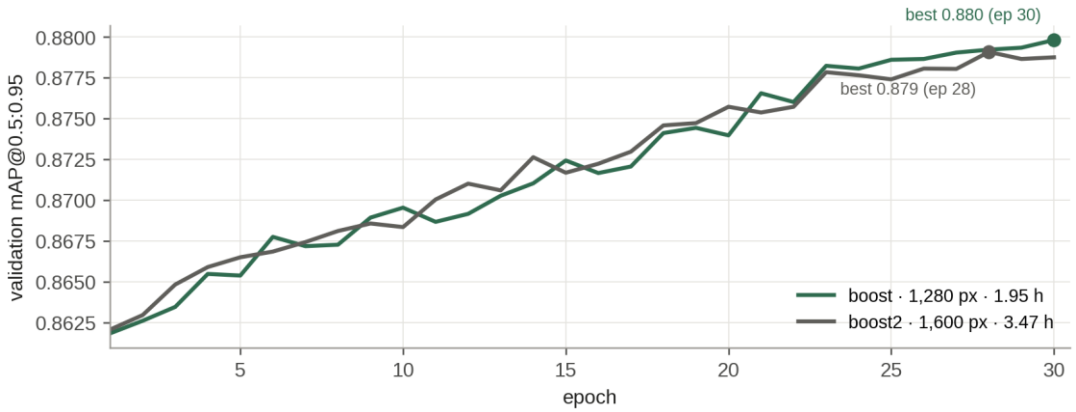


Figure 19. *Validation mAP@0.5:0.95 per epoch. Scores are lower than in Milestone 2 because the new validation set contains rotated grids, scenes and cut-outs.*

7.2 A rule sealed before the result

On 13 October, while boost2 was still training, the rule was written and its SHA-256 fingerprint recorded: *boost2 replaces boost if its mAP@0.5:0.95 on the 292 validation images is higher, each at its own image size; otherwise boost stays; thresholds are chosen on validation only; the test sets do not influence the choice.* The selection script refuses to run if the rule file no longer matches its fingerprint. boost scored 0.886 and boost2 0.878; in a paired bootstrap [1] boost2 was never higher in 1,000 resamples (Figure 20). boost stayed.

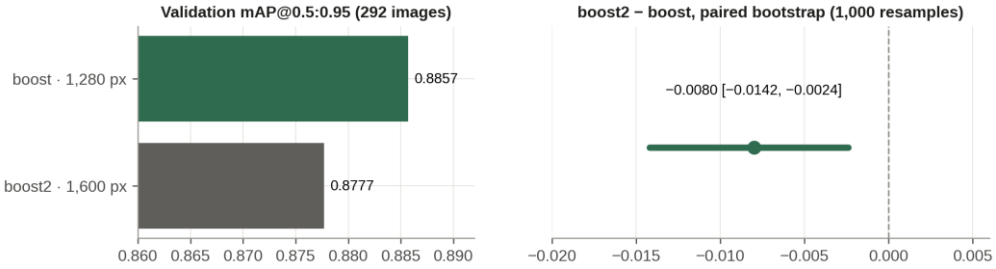


Figure 20. *The sealed decision: validation scores (left) and the paired bootstrap interval of the difference, entirely below zero (right).*

Thresholds were then chosen on validation. One threshold for all images was compared with separate thresholds for grids and scenes, to be adopted only if F1 rose by more than 0.003; it did. Grids use

confidence 0.10 with one box per 224-pixel cell; scenes, which have no cells, use 0.35 after class-agnostic duplicate removal at IoU 0.5.

7.3 Final results

Only after these choices were the test sets scored. boost improves on the Milestone 2 model on every test set (Figure 21, Table 9); every recall gain has a paired 95% interval that excludes zero. The largest gains are exactly where the diagnosis pointed: rotated grids (+16.6 points) and scenes, and above all cut-outs (+51.3 points), where the old model, deprived of its photos' own backgrounds, had found fewer than four objects in ten. Wrong IDs fell from 230 to 58 on rotated grids and from 16 to 10 on normal grids. The one measure that did not clearly improve is precision on cut-outs and single photos: the old model found few cut-out objects, but what it found was mostly right.

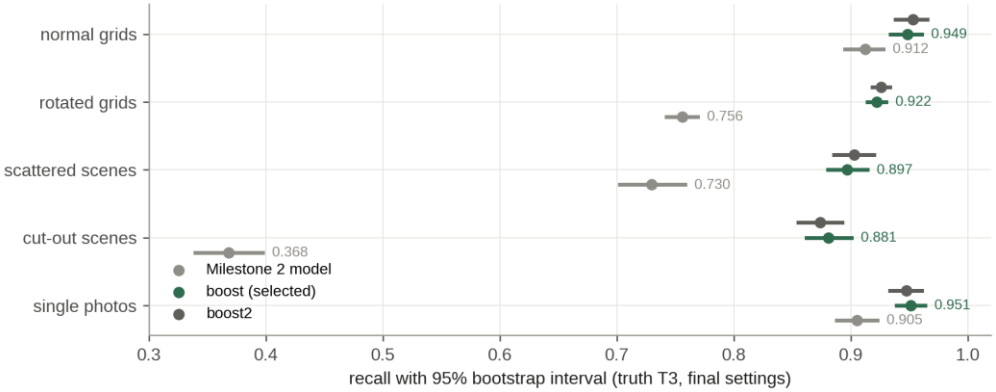


Figure 21. *Recall with 95% bootstrap intervals on the five test sets (T3, final settings).*

Table 9. Final results on T3 at the final settings, with 95% intervals.

Test set	Recall, M2 model	Recall, boost	Difference (points)	Precision	Fully correct
Normal grids	0.912 [0.895, 0.928]	**0.949** [0.934, 0.961]	+3.6 [+2.3, +5.0]	0.970	47.2% [36.1, 58.3]
Rotated grids	0.756 [0.743, 0.769]	**0.922** [0.914, 0.930]	+16.6 [+15.3, +18.0]	0.947	34.7% [28.2, 41.2]
Scattered scenes	0.730 [0.702, 0.759]	**0.897** [0.880, 0.914]	+16.8 [+14.0, +19.4]	0.943	25.0% [17.3, 33.7]
Cut-out scenes	0.368 [0.339, 0.397]	**0.881** [0.862, 0.901]	+51.3 [+48.0, +54.6]	0.914	28.8% [21.2, 37.5]
Single photos	0.906 [0.888, 0.923]	**0.951** [0.939, 0.964]	+4.6 [+3.1, +6.0]	0.968	95.1% [93.9, 96.4]

Precision and fully correct are boost's. boost's mAP@0.5 is 0.971, 0.953, 0.937, 0.912 and 0.970. The full table, with the Milestone 2 model and boost2, is in Appendix B.

7.4 boost2 in hindsight, and why training stopped

At the final settings boost2 is marginally better than boost on some test sets (normal-grid recall 0.953; 33.7% fully correct scattered scenes against 25.0%) and worse on others (cut-outs 0.874, single photos 0.948), always within the intervals. The sealed rule chose on validation and the test cannot reopen that choice; boost is also faster (36 against 54 ms per 3x3 grid) and needs 63% of boost2's computation. The experiment itself had a design flaw worth admitting: boost2 changed the image size, the labels and the batch size at once, so its result cannot be attributed to magnification alone. Training stopped after boost because the remaining options offered little that 72 test grids could show: boost had levelled off, a larger YOLO26-m would double the training time on a GPU already at its limit for an expected one or two points, and a

threshold per object would fit 73 settings to 292 validation images. Combining models was the one option worth testing properly; Section 9 does that.

8. Checking Our Own Results

A final report should be able to say how far its own numbers can be trusted. A second measurement run, using the same weights, test sets and scoring code (every reproduction matched the saved results to four decimals), answered the questions the milestone reports left open.

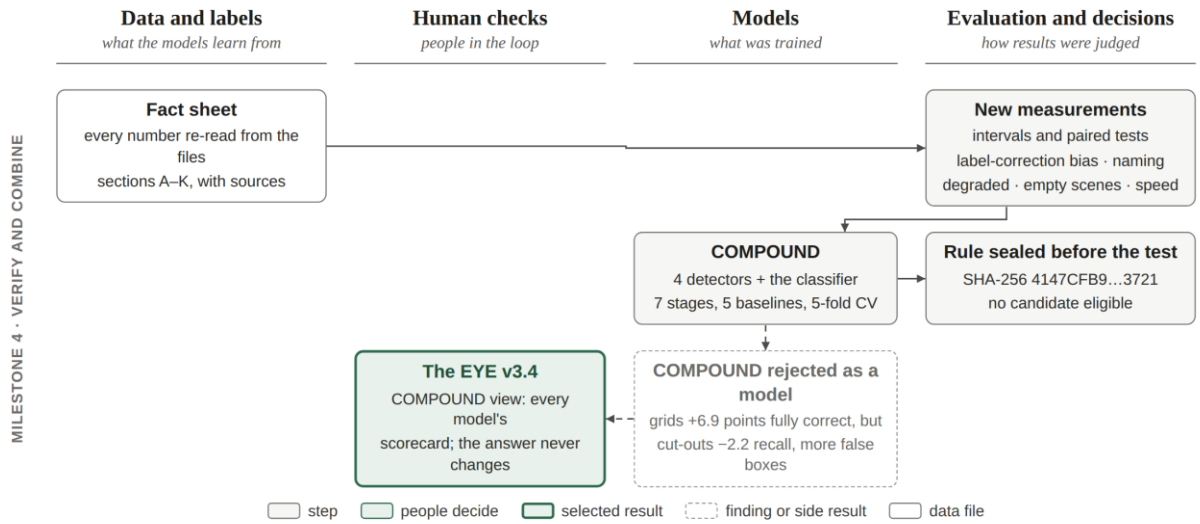


Figure 22. *Map of the verification for this report: the fact sheet, the new measurements and the COMPOUND experiment (Chapters 8 and 9).*

8.1 Did correcting the test labels favour boost?

Pass 2 of the label check used boost's disagreements to find label errors, and of its 17 test-box fixes the new box is closer to boost's prediction in 14. People made every decision, but the answer key moved towards boost. Scoring one set of predictions per model against all three answer keys measures the effect (Figure 23): boost's lead in recall on normal grids grows from 3.17 points on T1, which Milestone 3 never touched, to 3.82 on T3, and in mAP@0.5 from 1.5 to 2.4 points. The conservative, bias-free gain on normal grids is therefore recall 0.912 → 0.943 and mAP@0.5 0.940 → 0.955. On the other test sets the gains are far too large for a bias of this size to matter.

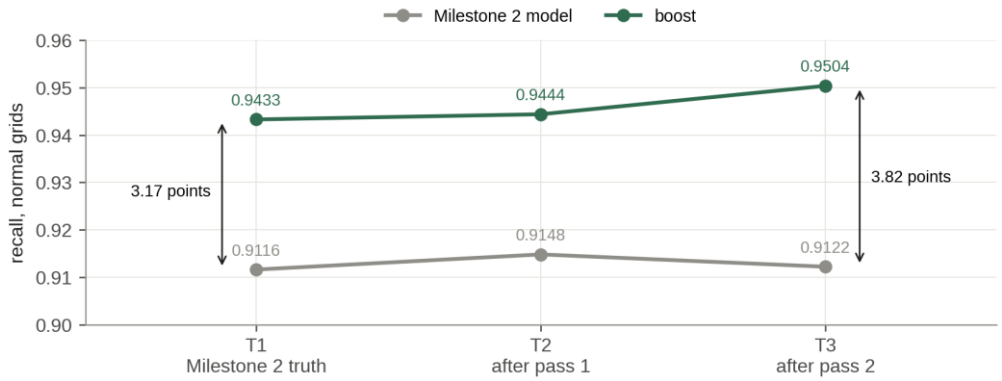


Figure 23. *Normal-grid recall of both models against the three answer keys.*

8.2 Does the detector name objects as well as the classifier?

Comparing 99.23% classification accuracy with detection recall mixes two tasks. The fair question, asked of both on exactly the same 1,040 single test photos, is whether the top prediction names the right object, ignoring the box. The classifier still names best: 99.5%, against 97.2% for boost and 94.7% for the Milestone 2 model (Figure 24). It is right on 25 photos that boost gets wrong, and boost on only one that the classifier gets wrong. Part of that edge may be inflated, because the classifier was trained before the near-duplicates were removed, but the gap points to a concrete design: let a detector find each object and a classifier name it.

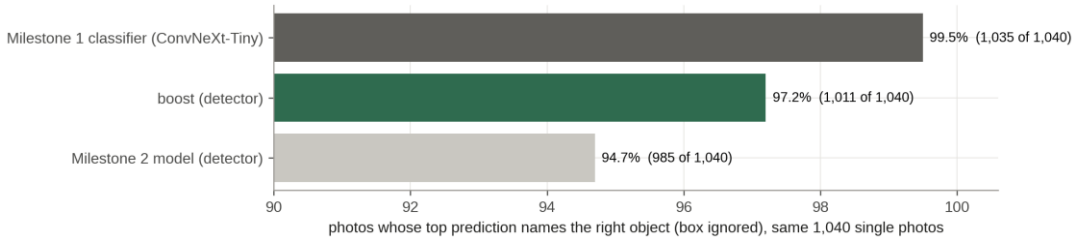


Figure 24. *Naming accuracy on the same 1,040 single photos.*

8.3 Degraded images and empty backgrounds

Two results run against the improvement. On the 72 degraded test grids boost is *worse* than the Milestone 2 model: mAP@0.5 0.393 against 0.465 and 156 wrong IDs against 65, with the project's scorer. Milestone 3's photo-damage augmentation was milder than this test, the likely but unproven reason. And on 100 background photos it never trained on, boost puts a box on 35% of them, the Milestone 2 model on 14%; a box names the photographer's own object in only 6% and 4%. The final detector rarely mistakes a room for its owner's object, but it is more eager to see something where nothing is.

8.4 Which objects gained

Per object (Figure 25), 42 objects gained in AP@0.5:0.95, 26 lost and 5 did not change. The largest gains went to the earlier problem objects: the keyboard (+0.23), the paddle (+0.21; AP@0.5 0.58 → 0.94, the object its label rule had not helped), the phone and the analog watch. Most losses are small and within the noise of 13 to 17 photos per object; the largest are a spoon (−0.11) and a pencil case (−0.07).

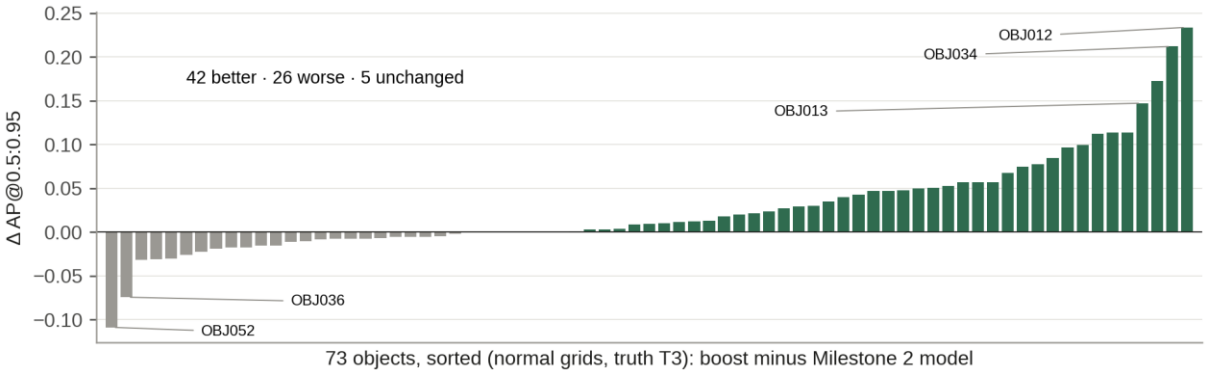


Figure 25. *Change in AP@0.5:0.95 per object, boost minus the Milestone 2 model (normal grids, T3).*

8.5 The stress test, speed and size

The Milestone 3 report quoted a stress test of 100 random 3x3 grids without saved per-grid data. Rerun with the same seeds and the EYE's own code, it reproduced the report exactly: 858 of 900 objects correct (95.3%) and 65.0% of grids fully correct (95% CI 55–74%), at 38 ms per grid. All three detectors are the

same network with 10,004,366 parameters; only the input size differs, and with it the cost: 72, 95 and 152 GFLOPs and 34, 36 and 54 ms per 3x3 grid on the laptop GPU (Figure 26).

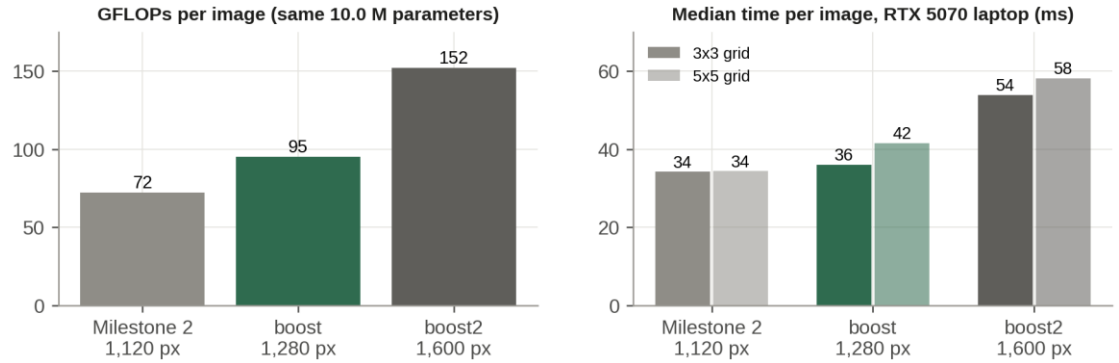


Figure 26. *Cost of the three detectors. Every input is rendered on a square canvas, so a 3x3 and a 5x5 grid cost almost the same.*

9. Can the Best Models Be Combined?

Each model had a measured strength: boost on rotation and scenes, the Milestone 2 model on degraded test images, a YOLOv8-s (Run 3) as a different architecture, and the classifier as the best namer of single photos. An insect's compound eye merges many small eyes into one picture; COMPOUND tested whether the project's models could be merged the same way. Because many settings could be tuned, the experiment was designed so that it could fail honestly.

9.1 Design

Every member's raw predictions were cached once on validation data only: the 292 validation images, a new degraded copy of the 68 validation grids made with the exact Milestone 2 degradation code, and the 53 validation-side backgrounds. Seven stages were added one at a time (Figure 27): orientation correction, weighted box fusion [16] of the four detectors, per-model calibration by temperature scaling [3], naming by the classifier over four rotations, gap filling when three members agree, quality weighting on poor images, and a logistic arbiter deciding each object. Five simpler baselines were scored alongside. Every setting was chosen on four folds of validation and scored on the fifth, from short lists fixed in advance; the teammates' runs were excluded because they had trained on our test photos.

The rule (SHA-256 4147CFB9…3721) made a candidate eligible only if, on validation, fully correct images rose by at least 2 points over boost alone, recall fell by no more than 0.5 points on any kind of validation image or on the degraded grids, and false boxes on empty backgrounds and empty filler cells did not increase. Its text was fixed in the instructions before the run; the rule file and fingerprint were written after one-line validation summaries of a discarded first run had been seen, while the code was unchanged from before both runs. The decision was recorded before any test prediction existed.

9.2 Results, and why it failed

Table 10. COMPOUND candidates on validation (out-of-fold).

	What the candidate adds	Fully correct	Cut-out recall
B0	boost alone at the final settings	46.2%	0.903
B1	boost with horizontal-flip test-time augmentation	47.6%	0.899
B2	boost with classifier renaming in grids	44.5%	0.903

	What the candidate adds	Fully correct	Cut-out recall
B3	boost and boost2, fused	52.0%	0.897
B4	boost and Run 3, fused	40.4%	0.890
C1	orientation correction	49.7%	0.905
C2	fusion of all four detectors	48.6%	0.876
C3	+ calibration	51.7%	0.878
C4	+ classifier naming	52.4%	0.882
C5	+ gap filling	52.4%	0.880
C6	+ quality weighting	53.1%	0.882
C7	+ arbiter	53.1%	0.882

B0's thresholds were chosen on this validation set, so B0 is scored in-sample, which slightly favours it.

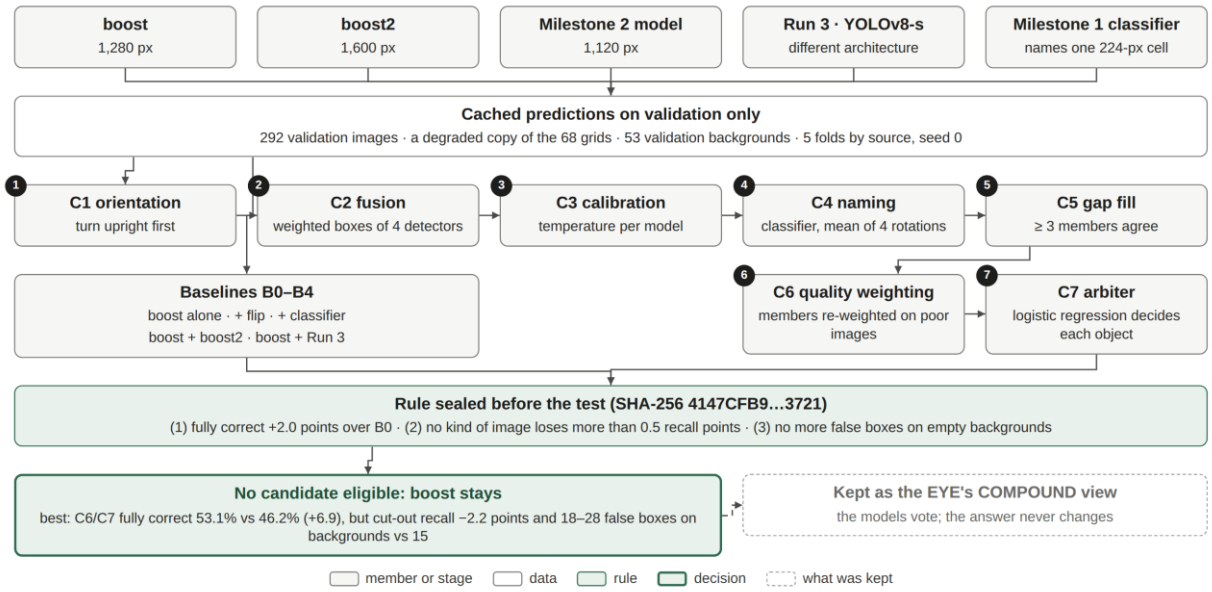


Figure 27. *The COMPOUND experiment: members, cached validation data, seven cumulative stages and five baselines, the sealed rule and the outcome.*

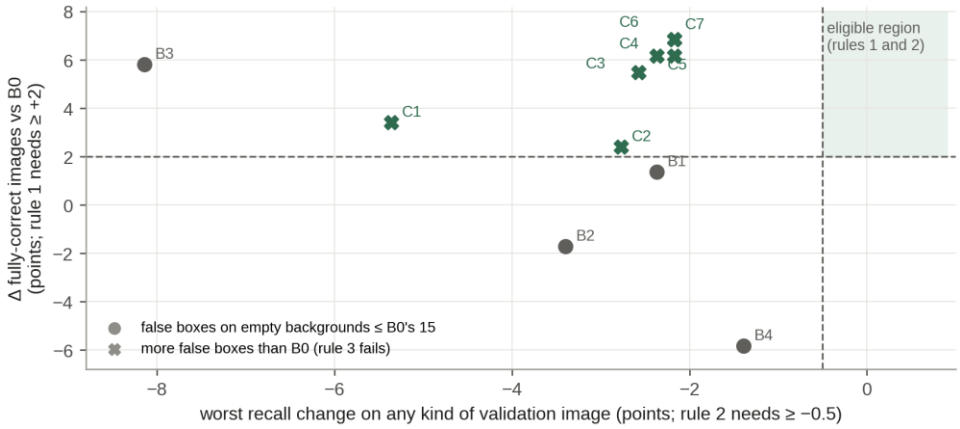


Figure 28. *Every candidate against the sealed rule: several clear rule 1, none clears rule 2, and the full combinations also fail rule 3 (crosses).*

No candidate was eligible, so boost stays. The failure is more instructive than a narrow win would have been. Combination helps grids and hurts scenes: the best candidates raised fully correct validation images from 46.2% to 53.1% and recall on degraded grids by 7.7 to 13.1 points, but fusion merged neighbouring objects in crowded scenes (cut-out recall −2.2 to −2.8 points) and let weak lone boxes through on empty backgrounds (18 to 28 false boxes against 15). Orientation correction failed for a reason worth remembering: it chose the right angle for only 37 of 68 rotated grids and turned 29 of 68 upright grids sideways, because boost, trained on rotated photos, is almost equally confident at every angle. The robustness Milestone 3 added removed the signal the trick relied on. And an intuition drawn from the degraded *test* set did not hold on validation, where the learned weights favoured Run 3 and boost rather than the Milestone 2 model.

The obvious next idea, combining models on grids only, would probably pass. It was suggested by these validation results, on a validation set now used four times, so testing it here would be selection after the fact; it is left as a hypothesis for new data. What survives is the explanatory value of the members' agreement, which the EYE now shows (Section 10).

10. The EYE: Making the Detector Explainable

A test score says how often a detector is right, not why it fails on a particular object. The EYE (Explainable YOLO Explorer) was built in Milestone 2 to answer that question one object at a time. It became the project's debugging tool, the source of most of the failure analysis above, and its public face.

10.1 Design

The EYE is a single Python file that runs a small web server; the page draws an eye whose iris is the DETECT button and whose pupil holds the image (Figure 29). Its central design decision is that it judges exactly like the evaluation. Random grids are built only from held-out test photos with person-checked boxes, or taken from the evaluation's own test scenes; each model runs at its own input size with its own validated threshold and duplicate rule; and every object is scored against the truth at IoU 0.5. A miss in the EYE is therefore a miss the report counts, which is why its stress test could reproduce the Milestone 3 report exactly. Figure 30 shows the detector it runs.

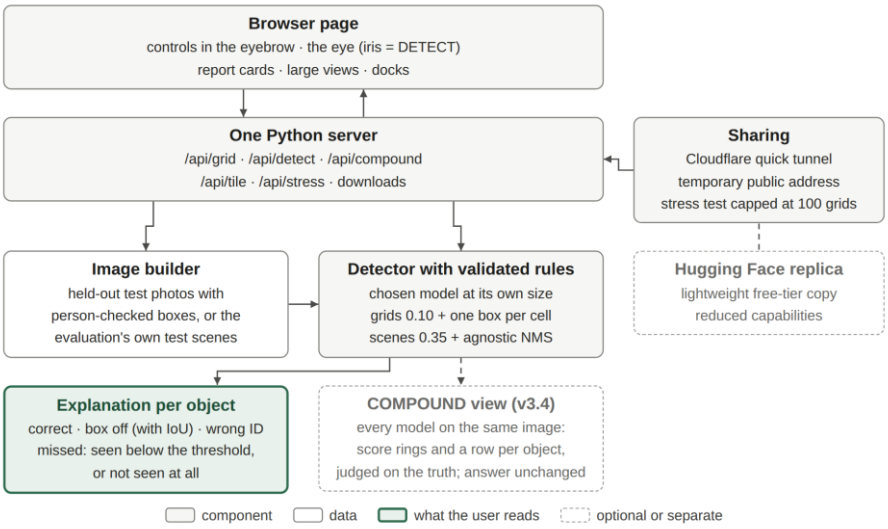


Figure 29. *Architecture of the EYE: one server builds the image, runs the chosen detector with the evaluation's rules and explains every object; v3.4 adds the COMPOUND view.*

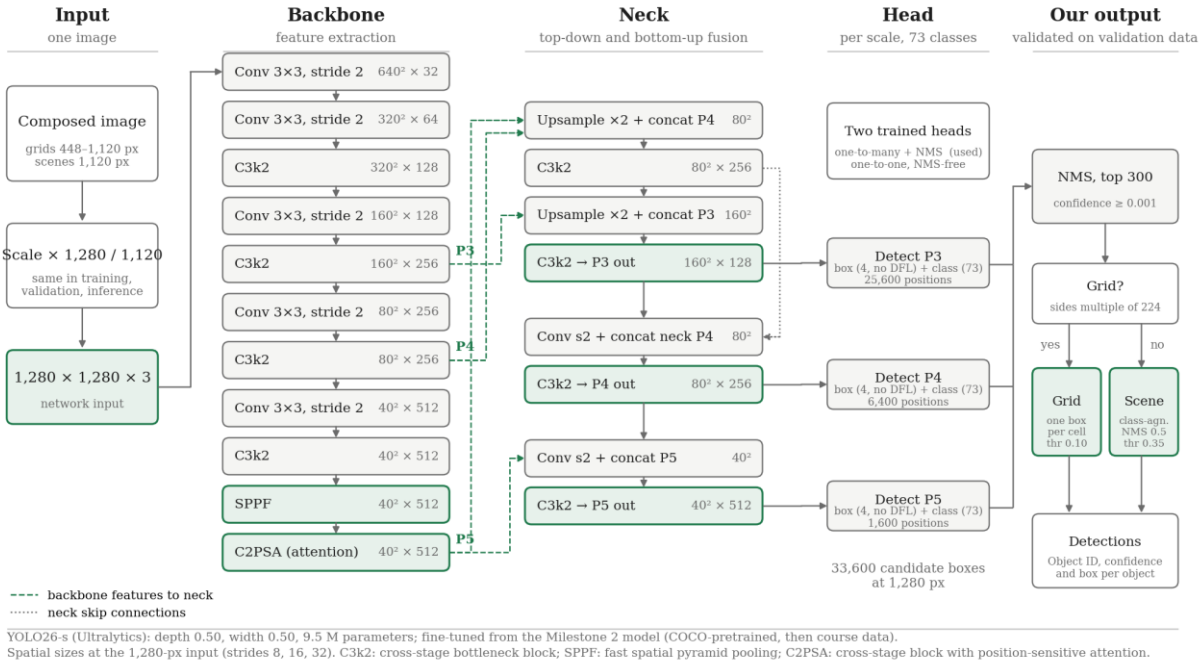


Figure 30. *The detector inside the EYE: YOLO26-s at 1,280 pixels with the project's post-processing (drawn for Milestone 3, which quoted 9.5 M parameters from an Ultralytics summary; the saved weights hold 10,004,366).*

10.2 How it explains

For every true object the EYE reports *correct*, *box off* with the overlap achieved, *wrong ID* with the look-alike it was confused with, or *missed*. For a miss it also inspects the model's weaker candidates below the threshold, so it can say whether the object was seen but not confidently enough, or not seen at all (Figure 31). That distinction separated threshold problems from perception problems and pointed Milestone 3 at the right fixes. From version 3.3, pressing DETECT moves the view into the pupil, where the image fills the screen with the score on one side and the results on the other (Figure 32).

10.3 From version 1 to 3.4

Table 11. Versions of the EYE.

Version	Port	What changed
1	7860	Milestone 2: random test grids with the evaluation's rules, an explanation per object, Inspect beside training photos, threshold sweep, stress test, downloads
2	7861	the first retrained model (boost) beside the Milestone 2 model
3	7862	every model and layout (grids, rotated grids, scattered scenes, cut-outs) with the validated thresholds; drag-and-drop upload; box-tightness view; the version behind the shared link
3.1–3.2	7863–7864	a calmer first screen and a close button on every window
3.3	7865	on DETECT the view moves into the pupil and the verdict appears beside it; sound; analysis of uploaded photos
3.4	7866	the COMPOUND view

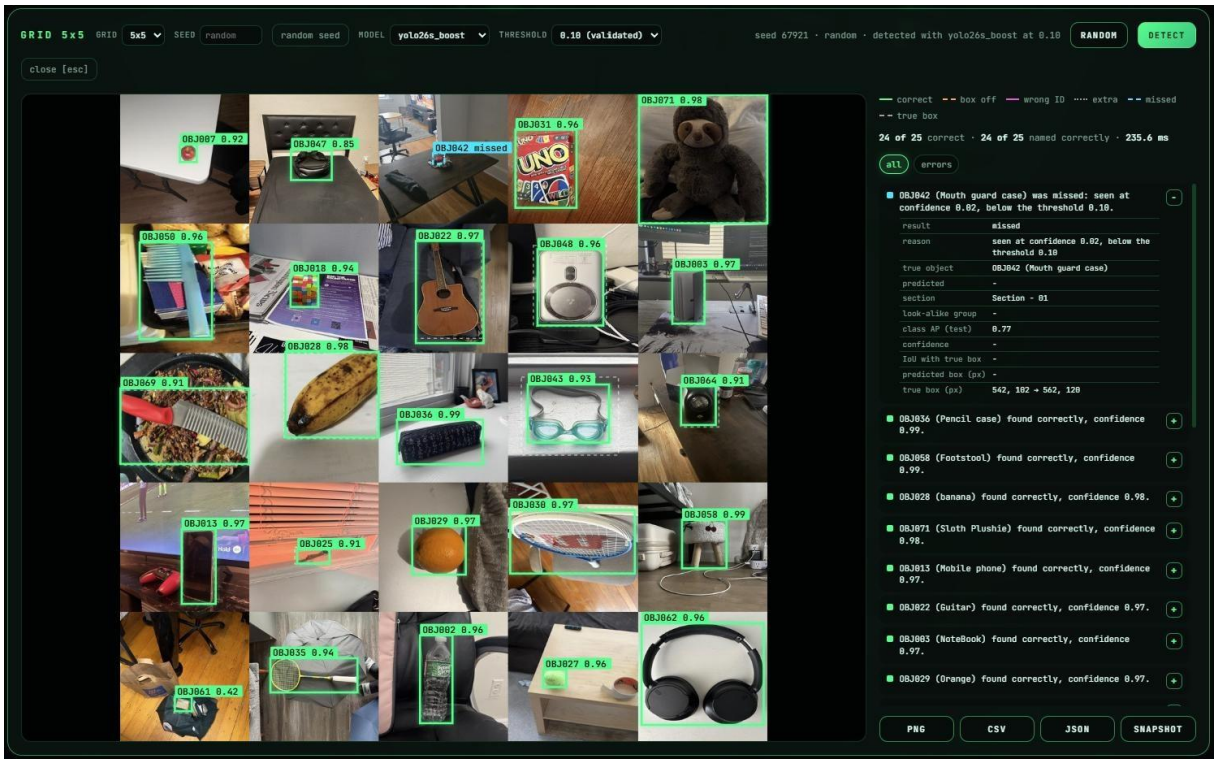


Figure 31. *A real explanation in the EYE: 24 of 25 objects correct; the mouth-guard case was missed, but seen at confidence 0.02, below the threshold 0.10.*

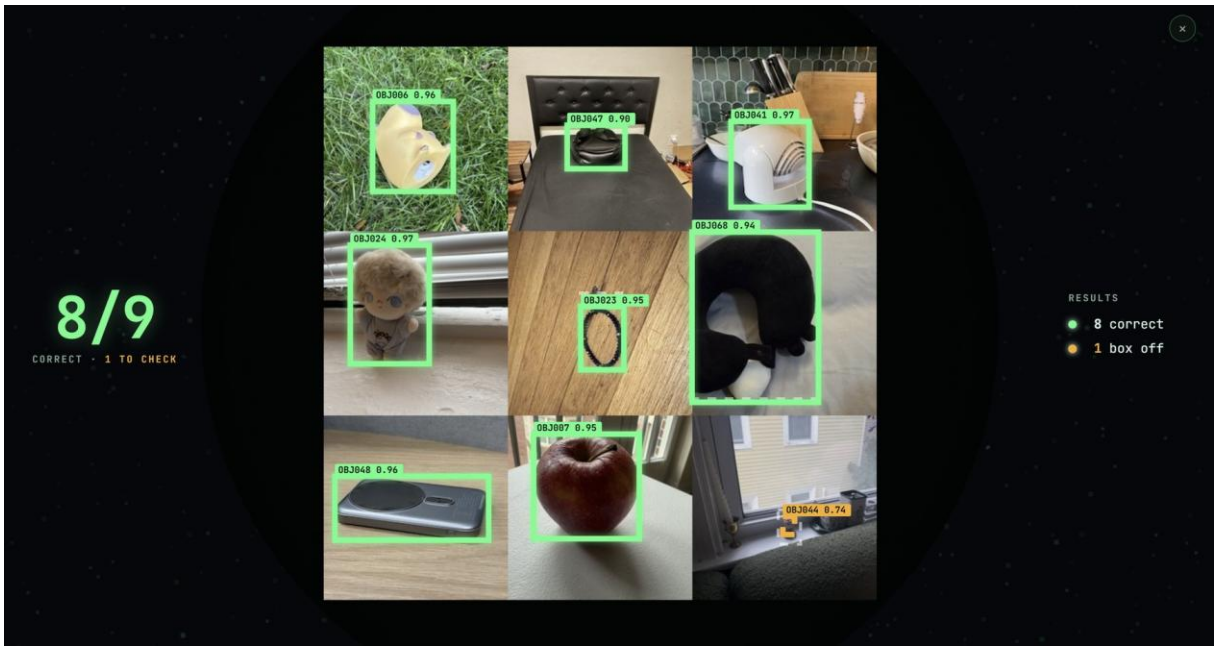


Figure 32. *The verdict view of version 3.3 on a 3x3 test grid: eight objects correct and one to check, the anime figurine, whose box is placed off.*

Version 3.4 turns COMPOUND's negative result into something useful. A switch runs the other models on the same image, each with its own validated settings, and shows a scorecard: on images with true boxes every model is judged against the truth exactly like the answer, as a score ring per model and one row per true object (correct, box off, named something else, or missed), with the objects on which the models

disagree listed first. The Milestone 1 classifier names the 224-pixel cell under each object. On uploads, which have no truth, the other models are compared with the answer's boxes instead. Nothing in the view changes the answer or the score: in an installation test on a 5x5 test grid, the stored detection table was byte-identical before and after.

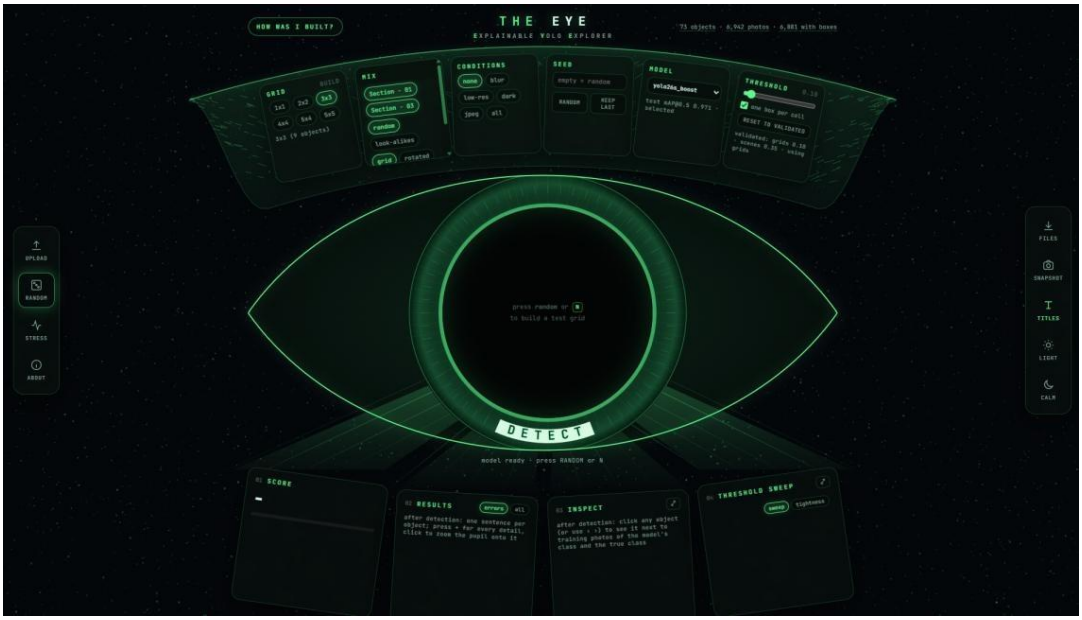


Figure 33. *The EYE v3: controls along the eyebrow, docks at the sides, the iris as DETECT button and four report cards below.*

That test also showed what the view is for. boost got 22 of the 25 objects right, boost2 23, the Milestone 2 model 22 and Run 3 21; every model got the same 21 objects right, and the four that separated them were small or easily confused objects. All four detectors missed the mouth-guard case and none boxed the power bank correctly, yet the classifier named all 25 objects, those two included. On one image the scorecard makes the case for a detector that finds and a classifier that names (Section 12).

10.4 Sharing it safely

The EYE needs the GPU and keeps images, detections and stress tests in memory, so it runs on the laptop that holds the models and photos. A free Cloudflare quick tunnel [18] gives it a temporary public address without accounts, firewall changes or code changes; scripts start it with a stress-test limit for visitors, keep the laptop awake, verify the address from outside, and later restore the laptop's exact power settings. Hosting the full EYE permanently was rejected for three reasons: Hugging Face Docker Spaces require a paid plan [19], free CPU hosting would make each detection take seconds, and the EYE shows photographs students took of their belongings in their rooms, which an always-on address would publish to anyone. A lightweight free-tier replica on Hugging Face, set up by Tanmayi, shows that the project can be deployed beyond the laptop; it has reduced capabilities and is not the version evaluated here.



Figure 34. *The COMPOUND scorecard on the same 3x3 grid: every detector found the anime figurine and named it correctly, but none boxed it tightly enough (IoU 0.28 to 0.36, against the 0.50 needed); the classifier named all nine objects.*

11. Discussion

11.1 Four threads across the milestones

Read together, the milestones tell more than any one of them. Four findings run through all three (Table 12). The clearest is the room. Milestone 1 showed that a classifier, which must always name an object, falls back on the room when there is none. Milestone 2's detector, tested without its photos' own backgrounds, found fewer than four cut-out objects in ten. Milestone 3, trained on real but unseen backgrounds, found nearly nine in ten, at the price of more boxes on empty scenes (Figure 35). A hypothesis raised in the first milestone was confirmed in the second and largely repaired in the third.

Table 12. Four threads followed through the project.

Thread	Milestone 1	Milestone 2	Milestone 3 and this report
Reliance on the room	empty rooms named as the photographer's object 35.7% of the time	cut-out recall 0.368	cut-out recall 0.881; more boxes on empty backgrounds (35% vs 14%)
Low-quality photos	the hardest condition (best 51.7%)	degraded grids mAP@0.5 0.428	boost worse than its predecessor: not repaired
Look-alikes	sunglasses and dark devices confused	paddle, card wallet and watches weakest	paddle AP@0.5 0.58 → 0.94; still among the remaining errors
Label consistency	folder labels 98.3% consistent	tissue box AP 0.30 → 0.995 from one rule	every box checked; the commonest upright miss had often been a label problem

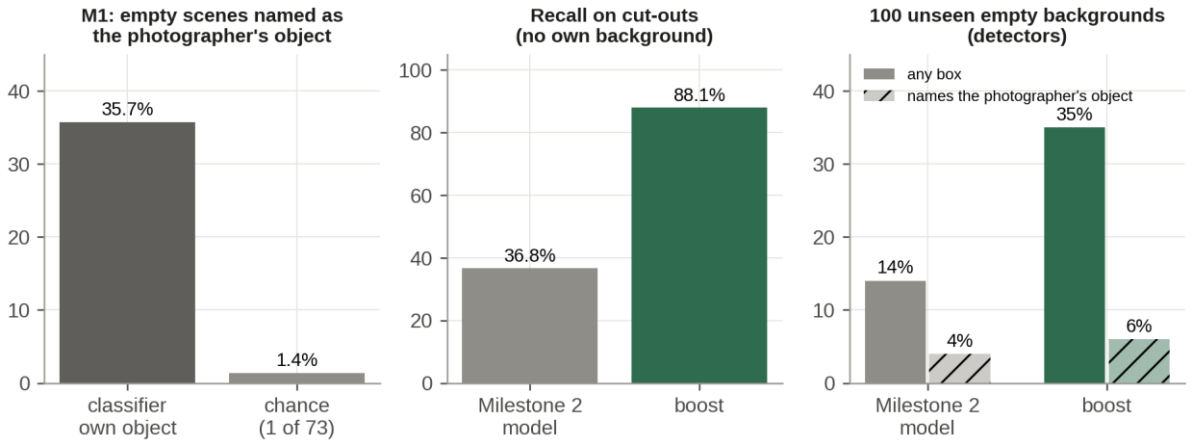


Figure 35. *The room thread across the milestones.*

11.2 How far can the results be trusted

The direction of the main results is secure; the size of the smaller effects is less so. Table 13 lists the threats we could identify, with their size where it could be measured. Two deserve emphasis. The same test photos served all three milestones; every choice was made on validation, but what the team saw in the EYE, which runs on test photos, shaped the direction of Milestone 3, a mild form of adaptive overfitting that sealed rules can limit but not remove. And the final answer key was partly corrected with the final model's help, an effect now measured at about 0.6 recall points on normal grids.

Table 13. Threats to validity.

Threat	Why it matters	Size or mitigation
Test photos reused across milestones	test errors seen in the EYE shaped later work	choices on validation; sealed rules; fresh random stress grids reproduce
Test labels corrected with boost's help	the answer key moved towards boost	measured: 0.6 recall and 0.9 mAP@0.5 points; T1 comparison reported
Small test sets	fully correct rates on 72 grids carry ±11-point intervals	intervals everywhere; claims limited to differences that exclude zero
Synthetic scenes	scattered and cut-out scenes are composites	unseen backgrounds; claims limited to these conditions
One split	another split could shift small differences	the same split keeps comparisons fair; replications agree
Classifier and near-twins	the classifier trained before 41 near-duplicates were removed	hard-subset accuracy 99.15%; naming comparison stated with this caveat
Lost raw outputs	the Milestone 1 main study cannot be re-read	a fully logged replication reached the same conclusion
Three scorers	Ultralytics and the project's two scoring codes differ slightly	scorer named per result; up to 0.7 points of mAP@0.5

The system is also limited in scope. It knows only these 73 objects and expects at most one per photo; it was trained on one person's photos of each object, on one laptop; it does not handle heavily degraded images; and real clutter, occlusion by other objects and motion blur were never tested.

11.3 Ethics and privacy

One photograph in the dataset is a phone screenshot showing personal details. Milestone 1 blurred it in every figure, together with twelve images an automatic screen flagged as possibly showing a phone screen; Milestone 2 removed it from all detection data, and it appears nowhere in this report. More generally the dataset shows students' belongings in their homes, which is why the full EYE is shared only through a temporary link within the course; wider publication would need the permission of everyone who contributed photos. The system is a course demonstrator for 73 known objects and is unsuitable for surveillance, safety-critical use or identifying people.

11.4 What we would do differently

- **Check every box from the start.** The full label check of Milestone 3 found problems that the vote's measured 97–99% accuracy could not rule out; doing it in Milestone 2 would have caught the inconsistent objects before training rather than after it.
- **Change one thing at a time.** boost2 changed image size, labels and batch size together, so its result cannot be attributed.
- **Correct labels without the candidate model's help,** or with two independent second opinions, to keep the answer key neutral.
- **Keep a final test set untouched until the last milestone,** so the final number is an unbiased estimate and not only a fair comparison.
- **Save every output where it can be re-read, and build the fact sheet continuously** rather than at the end.
- **Train for the worst condition explicitly.** The photo-damage augmentation of Milestone 3 was milder than the degraded test, and the model got worse there.

12. Conclusion and Future Work

The project began with a simple question, which object is in this photo, and ended with a detector that finds 73 objects in rotated grids, scattered scenes and cut-outs on backgrounds it has never seen, an explorer that shows why each object was found or missed, and a measured account of how far all of it can be trusted. ConvNeXt-Tiny recognised single photos at 99.23%; the Milestone 2 detector reached mAP@0.5 0.947 on grids; boost raised recall on every test set, most of all where the earlier model had leaned on its photos' backgrounds or had never seen rotation.

Three lessons carry beyond this dataset. Pretrained weights decide first. Consistent labels move results more than a better architecture: one rule moved the tissue box from AP 0.30 to 0.995, and checking every box was the most valuable step of Milestone 3. And only decisions made on validation, by rules fixed before the results, make a number worth reporting, including the numbers that say a promising idea did not work. COMPOUND is the clearest example: better on half the evidence and worse on the rest, rejected by the same standard that chose the final model, and kept where it helps, as a view that explains rather than decides.

The measurements of this report also point to the next steps. The classifier names single photos better than any detector, so a two-stage system in which a detector finds and a classifier names, trained on rotated and degraded photos, is the most promising measured opportunity. Combining models on grids only should be tested on new data under a new sealed rule. Real cluttered photographs, with several objects, occlusion and motion blur, would test what the synthetic scenes only approximate. Training with degradations at least as

strong as the test should repair the one thread this project did not close. And showing calibrated confidence together with the models' agreement would let a user of the EYE see when not to trust an answer.

Project repository: Github/TheEYE

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A. Models and Data

A.1 Model registry

Table 14. Every model of the project.

Model	Architecture	Trained on	Input	Key result	Weights · SHA-256 prefix
A2	custom 10-layer CNN, scratch	M1 train	224	89.06% test accuracy	Colab main study (outputs not saved)
C · D · G	ResNet-50 · EfficientNet-B0 · DenseNet-121	M1 train	224	98.66% · 98.75% · 98.66%	Colab main study
E (classifier)	ConvNeXt-Tiny, ImageNet	M1 train	224	99.23% (local replication 99.42%)	`milestone1_outputs_v2\…\E_convnext_tiny_finetune\best.pt` · 47AD2B85…
Box helper	YOLO26-s, COCO	3,915 training boxes	–	val mAP@0.5 0.950 (single photos)	Milestone 2 box pipeline
Runs 1–4	YOLOv8-s / YOLO26-s	M2 grids	1,120	test mAP@0.5 0.910–0.932	`milestone2_outputs\detect\runs\…`; Run 3 · 80B17F3F…
Run 5 (M2 model)	YOLO26-s, fine-tuned twice	M2 final grids	1,120	mAP@0.5 0.947	`…\runs\yolo26s_ft2\weights\best.pt` · 90C434C2…
boost (selected)	YOLO26-s from Run 5	3,000 robust images, pass 1	1,280	rotated recall 0.922, cut-outs 0.881	`M2 extra\boost\runs\boost\weights\best.pt` · 24FAA203…
boost2	YOLO26-s from Run 5	3,000 images, pass 2	1,600	not chosen (validation 0.878 vs 0.886)	`M2 extra\boost2\runs\boost\weights\best.pt` · D9970A44…
COMPOUND	ensemble of 4 detectors + classifier	–	–	rejected by the sealed rule	`Milestone 4\compound` (code, caches, rule)

All detectors are small ("-s") YOLO26 or YOLOv8 networks; SHA-256 prefixes identify the exact weight files used in this report's measurements.

A.2 Model cards

Table 15. Model card of the Milestone 1 classifier, after Mitchell et al. [13].

Field	Content
Intended use	Name which of the 73 course objects is in a single photo that shows one of them.
Not for	Photos with no object (it will still name one; 35.7% of empty scenes named as the photographer's object), objects outside the 73, people.
Training data	4,859 training photos, 224 px, ImageNet pretraining, one shared recipe (Section 3).
Evaluation	1,042 test photos: 99.23% (95% CI 98.7–99.7); hard subset 99.15%; 8 errors (4 multi-object scenes, 2 look-alikes, 1 very small object, 1 screenshot).
Robustness	Best of five models under every simulated problem; low-quality photos remain hard (best model 51.7%).
Caveats	Trained before near-duplicates were removed (99 test photos have a near-twin in training).

Table 16. Model card of the selected detector, boost.

Field	Content
Intended use	Find and name the 73 course objects in grids of their photos and in composite scenes, at any 90° rotation.
Settings	1,280 px; grids: confidence 0.10, one box per 224-px cell; scenes: 0.35 after class-agnostic NMS at IoU 0.5.
Training data	3,000 images built from training photos: rotated grids, scattered scenes on real backgrounds, SAM cut-outs, empty scenes; person-checked boxes (pass 1).
Evaluation (T3)	Recall: normal 0.949, rotated 0.922, scattered 0.897, cut-out 0.881, single photos 0.951; precision 0.914–0.970; intervals in Table 9.
Known weaknesses	Heavily degraded images (mAP@0.5 0.393, worse than the Milestone 2 model); boxes on 35% of empty backgrounds; tiny objects; look-alike groups.
Cost	10.0 M parameters, 95 GFLOPs, 36 ms per 3x3 grid and 42 ms per 5x5 on a laptop RTX 5070.

A.3 Datasheet and data files

Table 17. Datasheet for the 73-object dataset, after Gebru et al. [2].

Question	Answer
Motivation	Course dataset for comparing deep-learning methods on everyday objects photographed by students.
Composition	73 objects, 89–106 usable photos each; one object per photo by design; second objects sometimes visible (16 boxed in Milestone 3); separate background-only photos (385).
Collection	Each object photographed about a hundred times by one student, mostly in their own rooms: **object and background are correlated by construction**.
Preprocessing	Filename-based background removal, MD5 de-duplication, 224 × 224 resizing for models; no personal-data processing beyond removing one screenshot (Section 11).
Labels	Folder name = Object ID (audited, 98.3% agreement); boxes made in Milestone 2 by three box-drawers, a vote and people; every usable box checked by a person in Milestone 3.
Known biases	Background correlation; burst photos (near-twins); look-alike groups (watches, wallets, sunglasses, rackets, bottles); a few very small objects.
Recommended use	Closed-set recognition and detection of these 73 objects; not a benchmark for open-world detection or real clutter.

Table 18. Key data files.

File	Count	Content
`milestone1_outputs_v2\split.csv`, `clean_dataset.csv`	6,942	the cleaned photos and their split (seed 42)
`…\excluded_background_only.csv`	385	background photos removed by file name (378 distinct by content)
`…\removed_duplicates.csv`	16	exact duplicates removed, with the kept copy
`milestone2_outputs\boxes_master.csv`	6,936 rows	one box and status per photo (Milestone 2); never modified afterwards
`milestone2_outputs\detect\dataset\…`	1,500 / 68 / 72	Milestone 2 training images, validation and test grids with YOLO label files
`…\dataset_lowq`	72 grids	degraded copies of the test grids
`M2 extra\label_check\boxes_master_checked.csv`	6,881 usable	every box checked by a person (Milestone 3)
`…\label_check\decisions.csv`, `extra_boxes.csv`, `summary.json`	–	every review decision; 16 second objects; counts

File	Count	Content
`M2 extra\boost\…`, `boost2\tests\…`	3,000 / 292 / 5 sets	Milestone 3 training images, validation and test sets (T2, T3)
`Milestone 4\facts\M4_FACTS.md`, `measure2\`	–	the fact sheet and the new measurements behind this report
`Milestone 4\compound\`	–	COMPOUND code, caches, rule, provenance and results

B. Additional Results

Table 19. Milestone 1, fully logged local replication: test accuracy under each condition (same split; models A–F of the local run).

Model	Clean	Small	Crop	Occluded	Low quality
A Custom CNN	88.6	75.9	87.7	73.7	26.8
B ConvNeXt-T linear probe	97.4	85.7	96.9	89.4	50.6
C ResNet-50	98.7	95.6	98.0	91.7	41.8
D EfficientNet-B0	98.2	95.1	98.2	90.3	47.8
E ConvNeXt-Tiny ★	**99.4**	**97.9**	**99.0**	91.6	**57.0**
F ConvNeXt-T, basic augmentation	98.7	86.6	98.0	88.4	36.7

Model F isolates the augmentation: compared with E it lost 20.4 points on low-quality photos and 11.3 on small objects, evidence that the richer augmentation recipe was worth it.

Table 20. Milestone 3 final results with boost2 (T3, final settings; 95% bootstrap intervals).

Test set	Model	Recall	Precision	Fully correct	mAP@0.5
Normal	M2 model	0.912 [0.895, 0.928]	0.950 [0.935, 0.964]	34.7%	0.947
	boost	0.949 [0.934, 0.961]	0.970 [0.958, 0.980]	47.2%	0.971
	boost2	0.953 [0.938, 0.965]	0.973 [0.962, 0.982]	50.0%	0.972
Rotated	M2 model	0.756 [0.743, 0.769]	0.845 [0.831, 0.859]	9.3%	0.813
	boost	0.922 [0.914, 0.930]	0.947 [0.939, 0.955]	34.7%	0.953
	boost2	0.926 [0.919, 0.934]	0.946 [0.938, 0.954]	37.0%	0.953
Scattered	M2 model	0.730 [0.702, 0.759]	0.921 [0.905, 0.937]	8.7%	0.803
	boost	0.897 [0.880, 0.914]	0.943 [0.931, 0.955]	25.0%	0.937
	boost2	0.903 [0.886, 0.920]	0.944 [0.929, 0.958]	33.7%	0.942
Cut-out	M2 model	0.368 [0.339, 0.397]	0.935 [0.903, 0.961]	0.0%	0.559
	boost	0.881 [0.862, 0.901]	0.914 [0.896, 0.932]	28.8%	0.912
	boost2	0.874 [0.855, 0.893]	0.926 [0.907, 0.942]	24.0%	0.904
Singles	M2 model	0.906 [0.888, 0.923]	0.954 [0.940, 0.966]	90.4%	0.941
	boost	0.951 [0.939, 0.964]	0.968 [0.957, 0.979]	95.1%	0.970
	boost2	0.948 [0.933, 0.961]	0.967 [0.955, 0.978]	94.8%	0.966

C. Code, Reproducibility and Safeguards

C.1 Every script on record

Table 21. Every project file on record, in the order it was used.

File	Milestone	Jobs
`Milestone1_Expert_CNN_Study_LOCAL.ipynb`	1	index, filename background rule, MD5 duplicates, split, label audit, five CNNs, robustness, McNemar, Grad-CAM, background test
`M2_Boxes.ipynb`	2	exclusions; OWLv2 text and image prompts; YOLO helper; vote; review tool (cell 12); `boxes_master.csv`
`M2_Detect.ipynb`	2	grid datasets and label files; baked augmentation; Runs 1–5; validation choice; test; per-class AP
`m2_cross_eval.py`	2	cross-evaluation of runs A–C against the selected model with masking
`m2_demo_app.py`	2	a Gradio demo of the detector (features not recorded)
`m2_eye_app.py`	2	The EYE v1
`m2_diagnose.py`	3	diagnosis on validation; rotations; miss causes; cheap fixes
`m2_label_check.py`	3	prepare, review (port 7870), apply, merge; both passes
`m2_boost.py`	3	build (3,000 images, SAM cut-outs, test sets), train boost and boost2, evaluate
`m2_boost_val.py`	3	sealed selection rule, paired bootstrap, thresholds, one look at test
`m2_eye_app_v2.py` … `m2_eye_app_v34.py`	3–4	The EYE v2 to v3.4 (COMPOUND view)
`start_share.ps1`, `stop_share.ps1`, `restore_power.ps1`	3	sharing through a Cloudflare quick tunnel
`fix_paths.py`	3	path repair after restructuring the folders
`make_space_bundle.py`, `upload_space.py`	3	prepared for a Hugging Face Space; not used for the full EYE
`measure2.py`, `compound.py`	4	new measurements and the COMPOUND experiment for this report

C.2 Environment and commands

Table 22. Environment (identical on the training laptop for every milestone).

Item	Version
Hardware	NVIDIA GeForce RTX 5070 Laptop GPU, 8 GB; Windows 11
Python · PyTorch · torchvision	3.14.2 · 2.11.0 (CUDA 12.8) · 0.26.0
Ultralytics	8.4.174 (detectors, SAM 2.1 base `sam2.1_b.pt`)
OWLv2	google/owlv2-base-patch16-ensemble via transformers 5.2.0
Randomness	split seed 42; COMPOUND folds and bootstraps `numpy.default_rng(0)`; stress grids seeds 1000–1099
Milestone 1 main study	Google Colab (T4 instructed); outputs not saved

- **Milestone 1:** run the local notebook top to bottom; outputs go to `milestone1_outputs_v2`.
- **Milestone 2:** `M2_Boxes.ipynb`, then `M2_Detect.ipynb`.
- **Milestone 3:** `python m2_diagnose.py`; `python m2_label_check.py prepare | review | apply`; `python m2_boost.py build`, then `train`; `python m2_boost_val.py`.
- **The EYE:** `start_eye_v34.bat` (port 7866; points the EYE at `Milestone 3\M3 HOST`); sharing with `start_share.ps1`.

C.3 Engineering safeguards

Table 23. Engineering safeguards.

Safeguard	What it did
Fingerprints of submitted work	SHA-256 of the Milestone 2 weights, results file, box file and original EYE recorded in `protected_hashes_before.txt` and re-checked after every phase; all still match
Sealed rules	boost vs boost2 and COMPOUND decided by rules whose SHA-256 was recorded before results; the scripts refuse to decide if a rule file changes
New files, never overwrites	the label check writes `boxes_master_checked.csv` beside the untouched original; Milestone 3 training starts from a copy of the selected weights
Resumable training	an interrupted run continues from its last epoch with the same settings; a finished run is never overwritten
Read-only measurement runs	for this report, listings of 92,187 and 154,971 project files were identical before and after the two new runs
One integrator	all files committed by one member, so one copy of the code and one dataset state exist
Path repair after restructuring	`fix_paths.py` rewrote paths inside working copies after the folders were reorganised; originals were checked before any update

D. Decision Log

Table 24. Key decisions, with the alternatives and the evidence.

Milestone	Decision	Alternatives	Evidence and outcome
1	Split once, before anything else	split per milestone	one test set throughout; fair comparisons
1	Choose the classifier by robustness among the best	highest accuracy only	four models tied on clean photos; ConvNeXt-Tiny robust everywhere
2	Boxes by three drawers and a vote	draw all by hand; one detector	accepted boxes measured 97–99% right
2	Make the image prompt one vote only	trust it fully	it agreed with the text prompt on only 47.9%
2	Label rules in two rounds	more training	tissue AP 0.30 → 0.995; paddle not helped
2	Select Run 5 on validation mAP@0.5:0.95	best test number	0.930 on validation; test used once
3	Diagnose on validation before retraining	retrain at once	rotation found; cheap fixes rejected
3	Check every box, writing a new file	trust the vote	6,881 checked boxes; original untouched
3	Train boost at 1,280 px	1,600 px first	small objects only 5 of 38 misses; half the time
3	Seal the boost/boost2 rule before results	decide after seeing both	boost chosen; bootstrap confirms
3	Two thresholds (grids 0.10, scenes 0.35)	one threshold	adopted because F1 rose by more than 0.003
3	Share by temporary tunnel	Hugging Face, Codespaces, static hosting	cost, GPU need and privacy
3	Stop training after boost	bigger model, ensemble	plateau, small gains, noise on 72 grids
4	Seal the COMPOUND rule; keep boost	adopt the best-looking candidate	no candidate eligible; kept as an EYE view

E. Errata: Corrections to the Earlier Reports

Table 25. Corrections and clarifications, with evidence.

Report	Statement	Correction and evidence
Milestone 2, §8.4	degraded grids mAP@0.5 0.506	0.506 belongs to Run 4; the selected Run 5 scores **0.428** (mAP@0.5:0.95 0.342) on the same set (run results files)
Milestone 2, §3.2	split 1,041 validation / 1,042 test photos	true for Milestone 1's split; photos with a box are 1,040 / 1,041 after the 6 exclusions (`boxes_master.csv`)
Milestone 2, run order	Runs 1–2 numbered YOLOv8-s first	YOLO26-s was trained first; numbering follows the report, not the clock (run timestamps)
Milestone 2, Table 5	label-rule AP before and after	the run and split behind these values are not recorded; they are kept as reported
Milestone 1, §2	OBJ004 background photos remain in the data	true for Milestone 1; Milestone 2 removed the five photos before any box was made (`M2_Boxes.ipynb`)
Milestone 1, Table 10	378 background photos tested	385 were removed; no file records why 378 were tested; most likely the 385 contain 7 byte-identical copies (385 − 7 = 378), removed by the Colab notebook before the background filter; the local run used 385 (32.5% own ID)
Milestone 3, architecture	9.5 M parameters	the saved weights contain 10,004,366 parameters (`measure2`, item 8); 9.5 M is an Ultralytics summary count
Milestone 3, stress test	858/900 objects, 65.0% grids, no saved data	confirmed exactly, now with per-grid data (`item7_stress_3x3_per_grid.csv`)

F. Traceability of Key Claims

Table 26. Where each headline number comes from.

Claim	Source
99.23% Milestone 1 accuracy	Milestone 1 report Table 6 (main study); replication `milestone1_outputs_v2\model_comparison.csv`
6,942 photos; split 4,859 / 1,041 / 1,042	`milestone1_outputs_v2\data_manifest.json`
Vote 3,172 / 2,935 / 829; accepted test boxes 98.9%	`fuse_summary.json`; `review_SC.csv`
Run 5 mAP@0.5 0.947, recall 0.918	`milestone2_outputs\detect\results.json`
Diagnosis numbers	`M2 extra\diagnosis.json`
6,881 checked boxes, 41 duplicates, 14 excluded	`M2 extra\label_check\summary.json`
boost 0.8857 vs boost2 0.8777; bootstrap	`M2 extra\boost2\selection.json`; rule SHA-256 `C6A67BD9…F8C0`
Final results and intervals	`Milestone 4\facts\measure2\M4_MEASURE2.json`, `item1_ci.csv`
Label-correction effect; 14 of 17 fixes	`measure2` item 2, `item2_pass2_changes.csv`
Naming 99.5 / 97.2 / 94.7%	`measure2` item 3
Degraded and background results	`measure2` items 4 and 6
COMPOUND decision	`Milestone 4\compound\COMPOUND_RESULTS.json`, `decision_table.csv`, `PROVENANCE.md`; rule SHA-256 `4147CFB9…3721`
EYE v3.4	`m2_eye_app_v34.py`, SHA-256 `197FC4B8…F7D4`
EYE v3.4 scorecard test (5x5 grid, seed 899456)	installation test on the project machine, 18 October: `m2_eye_app_v34.py`, SHA-256 `6EF33364…1013`

G. Glossary

Table 27. Terms used in this report.

Term	Meaning
IoU	intersection over union: overlap of two boxes divided by their combined area; ≥ 0.5 counts as a match
Precision · recall	share of kept boxes that are correct · share of true objects found correctly
AP · mAP	area under one object's precision–recall curve · its mean over the 73 objects
mAP@0.5 · mAP@0.5:0.95	mAP at IoU 0.5 · the mean over IoU 0.50, 0.55 … 0.95, which rewards tight boxes
Fully correct	an image with every object found and named and no extra box
Confidence threshold	boxes below it are discarded; validated per model and image kind
NMS	non-maximum suppression: keeps the best of overlapping boxes; class-agnostic NMS ignores the object name
Transfer learning · fine-tuning	starting from weights learned elsewhere (ImageNet, COCO) · continuing to train them on our data
Mosaic	augmentation that stitches parts of four training images into one
OWLv2 · SAM	open-vocabulary detector used to propose boxes · segmentation model used to cut objects out
Weighted box fusion	averages overlapping boxes from several models into one
Temperature scaling	rescales a model's confidences so they are comparable with other models'
Bootstrap · paired bootstrap	resampling the test images to estimate uncertainty · resampling both models on the same draws to test a difference
McNemar's test	compares two classifiers on the same photos using only the photos where exactly one is right
MD5 · perceptual hash	fingerprint of a file's bytes · fingerprint of how an image looks
Near-twin	a different photo that looks almost identical (burst shots)
T1 · T2 · T3	the three test answer keys (Table 1)
Seed	the starting number of the random shuffle; the same seed gives the same split